

The Convex Origin of Fixed Costs[†]

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Abstract

Are models with fixed and convex adjustment costs different? I show they are ex-ante equivalent under Fenchel duality and characterize the generic analytic mapping from a distribution G of heterogeneous fixed costs to the dual cost function C . Based on the closed form, I show that the cost C is generically convex due to selection over adjusters, and that it depends not only on the level of G but also on its dispersion — a mean-preserving spread lowers C everywhere. The duality provides a cross-walk across literatures: menu costs, lumpy investment, rational inattention, discrete choice, and control costs are readings of one primitive G , each cost specification the parametric image of an implicit distributional assumption. Menu-cost and rational-inattention models of price-setting are observationally equivalent at the extensive margin under additive separability ([Alvarez, Lippi, and Oskolkov, 2022](#)), with exponential G the explicit bridge to the binary Shannon rational inattention of [Matějka and McKay \(2015\)](#).

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1 Introduction

Macroeconomics has two canonical models of adjustment frictions: *fixed* costs, which make adjustment lumpy and threshold-triggered, and *convex* costs, which make it smooth and gradual. The two are treated as different theories of how micro lumpiness aggregates into macro smoothness, with sharply different predictions for monetary non-neutrality, the Phillips curve, and welfare. This paper shows they are *ex-ante equivalent*. When fixed costs are heterogeneous, drawn from a distribution G , the threshold rule and a smooth choice of adjustment probability φ paid for by a convex cost $C(\varphi)$ deliver the same expected payoff and the same adjustment probability; the two are Fenchel duals, with C the quantile integral of G . The realised action is binary in both formulations; what coincides is the smooth ex-ante extensive margin. The mechanism is selection: low-cost agents adjust first, so the marginal non-adjuster faces a higher cost draw than the last — increasing marginal cost *is* convexity. The object that distinguishes “lumpy” from “smooth” is not the type of friction but the distribution G from which fixed costs are drawn.

The equivalence yields three contributions. The first, and the paper’s central payoff, is a *cross-walk* between literatures that have developed in isolation: because the construction depends only on (G, w) with the surplus left abstract, one primitive G can be read five ways: menu costs, lumpy investment, rational inattention, discrete choice, and control costs each adopt a cost specification that is the parametric image of an implicit assumption on G (Proposition 4, Table 1). Uniform G is the quadratic adjustment/attention cost of [Khan and Thomas \(2008\)](#) and [Gabaix \(2014\)](#); exponential G is the binary rational-inattention cost of [Matějka and McKay \(2015\)](#), a generalised Kullback-Leibler divergence; power-law G an iso-elastic cost sharing the exponent of the Tsallis q -divergence ([Tsallis, 1988](#)); i.i.d. Gumbel shocks the multinomial logit of [McFadden \(1974\)](#) (Proposition 5). The sharpest instance is an observational equivalence established by [Alvarez, Lippi, and Oskolkov \(2022\)](#): menu-cost and convex-cost (information-friction) price-setting generate identical extensive-margin behaviour under the same G ; here an exponential G identifies that convex cost with the binary Shannon rational inattention of [Matějka and McKay \(2015\)](#), pinning down which point of that class the menu-cost model occupies (Section 3.4). The duality thus makes two parallel literatures complementary: [Alvarez, Lippi, and Oskolkov](#)’s hazard identification on the G -side and [Matějka and McKay](#)’s information-cost specification on the C -side become coordinates on one object. The reverse reading gives rational inattention’s abstract penalty a concrete resource-cost meaning. Here $C(\varphi^*)$ is the expected physical cost paid by adjusters, recoverable from cost-distribution data ([Oskolkov and Lippi, 2026](#)).

The second is the explicit formula behind the cross-walk between G and C and the

economic foundation for generically convex C . The duality itself is not new — it is the binary-choice analogue of the Williams-Daly-Zachary identity (McFadden, 1978, 1981), and the convex cost coincides with the §2.2 object of Alvarez, Lippi, and Oskolkov (2022). What is new is the construction $G \mapsto C$ in (3): a closed-form, environment-free quantile integral depending only on (G, w) , portable across applications (Theorem 1). Two structural results follow, both turning on what heterogeneity does to the friction. The deterministic fixed cost, every firm facing the same cost, is a knife-edge singularity: any heterogeneity, however small, restores strict convexity (Proposition 2). The lumpy model is thus the singular limit, attained only when the cost distribution degenerates to a point mass; the convex model is the generic case. Dispersion also has a price: holding the mean fixed, a mean-preserving spread of G lowers the entire cost function, so the deterministic fixed cost is not only the singular case but the most expensive one — concentration, not the level, of costs is what makes adjustment dear (Proposition 3). This also yields an independent partial-equilibrium smoothness channel, distinct from the general-equilibrium aggregation debate of Khan and Thomas (2008): under uniform fixed costs each firm’s problem is already convex in the probability representation, so aggregation is not the source of the smoothness.

The third makes the dynamic model tractable by smooth methods. Value-function differentiability is not itself a new regularity — the menu-cost tradition obtains it by smoothing the value function with idiosyncratic shocks — but the duality delivers it from the convexity of C alone (Theorem 2), together with a closed-form first-order condition $S = C'(\varphi)$ that replaces threshold-specific machinery and puts the dynamic model directly into FOC-based heterogeneous-agent solvers (Reiter, 2009; Auclert, Rognlie, and Straub, 2020). The same smoothness makes a fully non-linear global solution feasible across the entire spectrum, including the Golosov-Lucas endpoint that linearised methods cannot reach (Section 4.5).

The construction’s leading application is canonical New Keynesian pricing, and it delivers two results of its own. The setup is the cross-walk made concrete: a power-law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ traces a continuous spectrum from Calvo to Golosov-Lucas, a two-dimensional map of generalized power laws (local hazard exponent against the global tail of G) places the major price-setting literatures as coordinates on one primitive, and a single hazard-exponent estimate calibrates the menu-cost and rational-inattention readings at once (Alvarez, Lippi, and Oskolkov (2022)’s $\nu \approx 2$ reads as $\hat{\gamma} = 1$). This adds a dimension to Alvarez, Lippi, and Oskolkov’s one-dimensional hazard-exponent spectrum: the local exponent fixes one coordinate and leaves the tail of G free. The two results follow from the smooth construction. First, it makes the full dynamic model solvable by a single global non-linear method across the entire spectrum, the Golosov-Lucas corner included, which linearised methods cannot reach. Second, using that solution I show that the tail of G — the dimension the local

hazard slope, and even price-change kurtosis, leave unidentified — moves the selection dynamics, so it is a locally unidentified margin along which monetary transmission nonetheless varies (Section 4.5). The duality applies more broadly to any extensive-margin decision with heterogeneous costs (Section 5).

Related literature The analysis connects work on menu costs and pricing (Caplin and Spulber, 1987; Dotsey, King, and Wolman, 1999; Golosov and Lucas, 2007; Midrigan, 2011; Nakamura and Steinsson, 2010; Alvarez, Le Bihan, and Lippi, 2016; Vavra, 2014) and lumpy investment (Thomas, 2002; Gourio and Kashyap, 2007; Oskolkov and Lippi, 2026). The repricing hazard $\Lambda(x)$ realises the “adjustment function” of Caballero and Engel (1993, 1999), derived from G rather than assumed; the Vavra (2014) dispersion statistic is endogenous in G , and Caballero and Engel (2007)’s amplification result has an analogue in γ (Online Appendix C). Most closely, Alvarez, Lippi, and Oskolkov (2022) establish the same random-cost / convex-cost equivalence in a continuous-time sticky-price setting; Theorem 1 establishes it directly between G and C in discrete time, with an explicit closed-form construction portable across applications. The fixed-cost model is a control-cost model (Mattsson and Weibull, 2002; Hofbauer and Sandholm, 2002; Gabaix, 2014) with penalty set by G ; binary rational inattention (Sims, 2003; Matějka and McKay, 2015) yields KL, recovered here from exponential G , and the multinomial logit from i.i.d. Gumbel costs. Concurrent work by Bloedel, Denti, and Pomatto (2025) uses Fenchel duality with a different primitive — an information-acquisition cost rather than a distribution of physical fixed costs.

Roadmap Section 2 presents the duality theorem, characterises the induced cost function, establishes universality and the entropy correspondence, and reads one distribution G as five cost specifications. Section 3 works the cross-walk through New Keynesian pricing — the menu-cost–rational-inattention observational equivalence and the two-dimensional map. Section 4 reports a compact calibration and a global non-linear solution across the spectrum. Section 5 discusses further applications and scope; Section 6 concludes.

2 The duality

2.1 Primitives

Consider an agent who faces a binary decision: take an action (“adjust”) or maintain the status quo (“do nothing”). Adjusting yields a gross surplus $S \geq 0$ but requires paying a fixed cost $w\xi$, where $w > 0$ is a wage or unit cost and ξ is a random draw from a

distribution G supported on $[0, \bar{\xi}]$ with $\bar{\xi} \in (0, \infty]$ (the case $\bar{\xi} = \infty$ covers unbounded supports such as the exponential). We assume G has density $g > 0$ on its support and finite mean $E_G[\xi] < \infty$. All integrals against dG are taken over the support of G . Not adjusting yields zero, since the status-quo branch carries neither surplus nor cost. The randomness in ξ captures heterogeneity in the cost of adjustment. This heterogeneity may be across agents (a cross-section of firms facing different menu costs) or across time (a single firm drawing a new cost each period, as in [Dotsey, King, and Wolman 1999](#)).

This setup is deliberately minimal, stripping away the details of particular applications to isolate the structure they share: in the investment application, S is the surplus from investing at the optimal scale; in the pricing application of [Section 3](#), it is the gain from closing the price gap. Because the setup is general, the duality theorem applies to any extensive-margin decision where the agent's cost of acting is uncertain. The setup is, however, *additively separable* in the cost realisation: ξ enters only the contemporaneous payoff through $-w\xi$ on the adjust branch and not at all on the non-adjust branch, and in particular does not enter the continuation value of either branch.¹

Definition 1 (Fixed cost formulation).

Given the payoff S , the following defines the fixed cost payoff $\Pi^{FC}(S)$ and the threshold rule ξ^ : The agent observes the realised cost ξ and adjusts if and only if $w\xi \leq S$:*

$$\Pi^{FC}(S) = \int_0^{\bar{\xi}} \max\{S - w\xi, 0\} dG(\xi) = \int_0^{\min\{S/w, \bar{\xi}\}} (S - w\xi) dG(\xi). \quad (1)$$

The agent adjusts for all cost draws below the threshold $\xi^ = S/w$ and does not adjust otherwise.*

Definition 2 (Probability choice formulation).

Given the payoff S , the following defines the probability choice payoff $\Pi^{PC}(S)$ and the optimal adjustment probability φ^ : The agent chooses $\varphi \in [0, 1]$, paying a convex cost $C(\varphi)$:*

$$\Pi^{PC}(S) = \max_{\varphi \in [0, 1]} \{\varphi \cdot S - C(\varphi)\}, \quad (2)$$

where $C : [0, 1] \rightarrow \mathbb{R}_+$ with $C(0) = 0$, C increasing and convex. The agent trades off the expected benefit of adjusting ($\varphi \cdot S$) against the cost of choosing a high adjustment probability ($C(\varphi)$).

¹This rules out applications in which the realised cost depletes a stock that itself enters the continuation — e.g., household-side adjustment problems where ξ is paid from financial wealth that then affects the next-period marginal utility of consumption. [Section 5](#) discusses the consequences of relaxing this.

The duality theorem establishes that these two formulations are ex-ante equivalent: same value functions and same adjustment probabilities, with C derived from G via the quantile integral. For each distribution G with a positive density, there exists a unique convex cost function C with $C(0) = 0$ that delivers this equivalence.

2.2 Main result

Theorem 1 (Analytical convex duality of fixed costs).

The fixed cost formulation (Definition 1) and the probability choice formulation (Definition 2) are equivalent under the convex cost function constructed in closed form by the quantile integral

$$C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi). \quad (3)$$

Specifically:

(i) **Payoff equivalence.** $\Pi^{FC}(S) = \Pi^{PC}(S)$ for all $S \geq 0$. The two formulations are Fenchel conjugate duals: $\Pi^{PC}(S)$ is the conjugate of C , and $C(\varphi)$ is the conjugate of Π^{FC} .

(ii) **Policy equivalence.** The optimal adjustment probability is the same: $\varphi^* = G(S/w)$. Moreover, if the probability choice model randomises by drawing $\xi \sim G$ and adjusting whenever $\xi \leq G^{-1}(\varphi^*)$, the two models produce identical binary outcomes path by path.

The construction (3) is the static, environment-free analytical link between the primitive distribution G and the dual convex cost C : it depends only on (G, w) , with the surplus S left abstract.

Proof.

(i) **Payoff equivalence.** Substituting the threshold $\xi^* = S/w$ (assuming $S/w \leq \bar{\xi}$; for $S/w > \bar{\xi}$, both formulations yield $\Pi(S) = S - wE[\xi]$ at $\varphi^* = 1$, and for unbounded support $\varphi^* \in (0, 1)$ for every finite S) into (1):

$$\Pi^{FC}(S) = S \cdot G(S/w) - w \int_0^{S/w} \xi dG(\xi). \quad (4)$$

The first term is the expected surplus from adjusting: the probability of adjustment $G(S/w)$ times the surplus S . The second term is the expected cost paid by adjusters. Setting $\varphi = G(S/w)$ so that $S/w = G^{-1}(\varphi)$:

$$\Pi^{FC}(S) = \varphi \cdot S - w \int_0^{G^{-1}(\varphi)} \xi dG(\xi) = \varphi S - C(\varphi).$$

This is the objective of (2) evaluated at $\varphi^* = G(S/w)$. Since $C'(\varphi) = wG^{-1}(\varphi)$ is non-decreasing, C is convex, so the first-order condition $S = C'(\varphi)$ is sufficient and $\varphi^* = G(S/w)$ attains the supremum. The conjugate relationship follows from the definition of the Fenchel conjugate: $\sup_{\varphi} \{\varphi S - C(\varphi)\} = \Pi^{PC}(S) = \Pi^{FC}(S)$.

(ii) Policy equivalence. The first-order condition of (2) is $S = C'(\varphi)$. Differentiating (3) via the fundamental theorem of calculus gives $C'(\varphi) = wG^{-1}(\varphi)$, so $G^{-1}(\varphi^*) = S/w$, i.e., $\varphi^* = G(S/w)$: the same threshold probability as in Definition 1. For the path-by-path equivalence: in the fixed cost model, the agent draws $\xi \sim G$ and adjusts whenever $\xi \leq S/w$, which occurs with probability φ^* . If the probability choice model uses the same draw and adjusts whenever $\xi \leq G^{-1}(\varphi^*)$, the two rules coincide and the realised adjustment indicators are identical. The path-by-path coupling presupposes i.i.d. cost draws across time and agents; persistent-type heterogeneity in ξ preserves distributional but not trajectory equivalence. ■

In the fixed cost model, agents with low cost draws adjust and those with high draws do not. The expected payoff equals the adjustment probability times the surplus, minus the average cost paid by adjusters. In the probability choice model, the agent chooses an adjustment probability and pays a cost $C(\varphi)$. Proposition 1 below confirms that C is convex and characterises its shape. The theorem says these two descriptions yield exactly the same payoff and adjustment probability, provided C is constructed from G via (3).

The Fenchel conjugate structure places the two formulations on equal mathematical footing. The payoff function $\Pi^{PC}(S)$ and the cost function $C(\varphi)$ are conjugates of each other: the Fenchel conjugate of C yields Π^{PC} , and the conjugate of Π^{FC} yields C . The two express the same ex-ante value function in dual variables, one in the surplus S , the other in the adjustment probability φ . Any result proved for one variable translates immediately to the other. At the ex-ante value/probability level, the two are not different theories but different parametrisations of a single theory; under different primitive interpretations of G (physical cost versus information-processing penalty), the welfare and identification implications can differ, as discussed in Section 2.7. When G is uniform, C is quadratic and the equivalence can be verified by direct calculation (Corollary 2; Online Appendix B).

Relation to prior conjugacy results The binary conjugate identity is not new: it is the binary-choice case of the Williams-Daly-Zachary theorem (McFadden, 1978, 1981) (with cost draws $\xi \sim G$ in place of taste shocks, isomorphic under a sign change; the precise correspondence is in Online Appendix F), and in the menu-cost case Theorem 1's $C(\varphi)$ coincides exactly with w times the Section 2.2 convex flow cost of Alvarez, Lippi, and Os-

kolkov (2022): both equal $w \int_0^\varphi G^{-1}(s) ds$. What is new is twofold. First, the construction is static and environment-free, depending only on (G, w) with the surplus S left abstract, so the closed-form quantile integral $C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi)$ maps the primitive G to C directly and ports across applications (investment, rational inattention, labour supply) without re-derivation, whereas ALO recover the cost *from the observed hazard*, within a specific dynamic environment, by solving a second-order ODE. Second, Theorem 1(ii) constructs a coupling under which the realised binary outcomes coincide path by path, not merely in distribution — a statement absent from both WZD and ALO. In the game theory literature, Mattsson and Weibull (2002) and Hofbauer and Sandholm (2002) study agents who choose mixed strategies subject to convex “control costs” that penalise departures from a default action. Deviating from habitual behaviour is costly, and the cost increases as the deviation becomes more extreme. The probability choice formulation (2) has exactly this structure: $C(\varphi)$ is the cost of choosing adjustment probability φ relative to the default of not adjusting ($\varphi = 0$). Theorem 1 therefore shows that the fixed cost model *is* a control cost model. The control cost is not an ad hoc penalty imposed by the modeller, but the endogenous cost function determined by the distribution of frictions that the agent faces.

2.3 Properties of the induced cost function

The induced cost function $C(\varphi)$ inherits its shape entirely from the distribution G . The following proposition characterises its key properties and provides the economic interpretation that underlies all subsequent applications.

Proposition 1 (Cost function properties).

The induced cost function (3) satisfies:

- (i) $C(0) = 0$, $C(1) = w E_G[\xi]$.
- (ii) $C'(\varphi) = w G^{-1}(\varphi)$: *marginal cost equals the cost of the marginal adjuster.*
- (iii) $C''(\varphi) = w/g(G^{-1}(\varphi)) > 0$: *strictly convex whenever $g > 0$.*
- (iv) C is C^2 if g is C^1 and positive.

Proof. See Online Appendix A. ■

Part (ii) carries the core economic content, because it ties the marginal cost directly to the ordering of agents: agents are ranked by their cost draws, with those holding the lowest ξ adjusting first. The marginal cost $C'(\varphi) = w G^{-1}(\varphi)$ is the cost of the agent just indifferent at the current threshold: the φ -th quantile of the cost distribution. Each increment in φ

requires inducing a more expensive agent to adjust, which is the force that makes C convex. This is the economic origin of convexity: it is not assumed but arises endogenously from the ranking of heterogeneous agents by their costs.

Part (iii) shows the density g governs the strength of convexity: crowded margins (large g) give gentle curvature, sparse margins (small g) sharp curvature; the three monotone cases are in Corollary 1.

Corollary 1 (Shape of G determines curvature).

Since $C''(\varphi) = w/g(G^{-1}(\varphi))$ by Proposition 1(iii), the monotonicity of the density g across the support of G determines the monotonicity of C'' as φ traverses $(0, 1)$. The three canonical cases are:

- (i) *g increasing $\Rightarrow C''$ decreasing: there are relatively more agents at higher cost quantiles, so inducing additional adjustment becomes progressively easier. Adjustment is “cheap at the margin” for high φ .*
- (ii) *g decreasing $\Rightarrow C''$ increasing: agents thin out at higher cost quantiles, so inducing additional adjustment becomes progressively more expensive. Adjustment becomes disproportionately costly at high φ .*
- (iii) *g constant (uniform) $\Rightarrow C''$ constant: agents are equally spaced throughout the cost distribution, yielding the standard quadratic cost function with constant curvature.*

Proof. See Online Appendix A. ■

Corollary 2 (Canonical distributions).

Applying (3) to four canonical distributions recovers standard cost-function specifications in closed form:

- (i) **Uniform** $G = U[0, \bar{\xi}]$: $C(\varphi) = \frac{w\bar{\xi}}{2}\varphi^2$. *The constant density produces constant curvature, so each unit increase in the adjustment probability costs the same marginal increment. This is the standard quadratic adjustment cost used throughout the macroeconomics and industrial organisation literatures.*
- (ii) **Power law** $G(\xi) = (\xi/\bar{\xi})^\gamma$, $\gamma > 0$: $C(\varphi) = \frac{\gamma}{\gamma+1}w\bar{\xi}\varphi^{1+1/\gamma}$, *iso-elastic with elasticity $1 + 1/\gamma$. The shape parameter γ governs curvature: $\gamma < 1$ (cheap adjusters abundant) makes C more curved than quadratic, $\gamma > 1$ less; $\gamma = 1$ is uniform, $\gamma \rightarrow \infty$ linear (the degenerate limit), $\gamma \rightarrow 0$ free adjustment. The family spans lumpiness to frictionless adjustment and carries the pricing application of Section 3.*

- (iii) **Exponential** $G(\xi) = 1 - e^{-\lambda\xi}$ on $[0, \infty)$: $C'(\varphi) = -\frac{w}{\lambda} \ln(1 - \varphi)$, giving $C(\varphi) = \frac{w}{\lambda} [(1 - \varphi) \ln(1 - \varphi) + \varphi]$. The logarithmic marginal cost diverges as $\varphi \rightarrow 1$, severely penalising near-certain adjustment. This reflects the exponential distribution's unbounded support: to guarantee adjustment with probability close to one, the agent must be willing to pay arbitrarily high cost draws, which makes the marginal cost of the last increment unbounded, even though the total expected cost $C(1) = w/\lambda$ remains finite. Section 2.6 identifies this cost function with the generalised KL divergence of binary rational inattention.
- (iv) **Degenerate** at $\bar{\xi}$: $C(\varphi) = w\bar{\xi}\varphi$. The cost function is linear, with no strict convexity. The only optimal choices are $\varphi = 0$ (never adjust) or $\varphi = 1$ (always adjust), depending on whether S falls below or above $w\bar{\xi}$. This is the canonical binary threshold rule of the deterministic fixed cost model, the singular case analysed in Proposition 2 below.

Proof. See Online Appendix A. ■

These are not four different models: they are four special cases of one model, distinguished solely by the shape of G .

2.4 Universality of the convex representation

The degenerate case — a single, known fixed cost — is the only setting in which the problem is not strictly convex. This section shows that this case is a knife-edge singularity: arbitrarily close perturbations of a point mass by absolutely continuous distributions with positive density restore strict convexity. The convex representation is the non-knife-edge case; the deterministic fixed cost is the singular exception.

Proposition 2 (Universality).

Let $\bar{\xi} > 0$ be a deterministic fixed cost. For any $\epsilon \in (0, \bar{\xi})$, perturb the cost to $\xi \sim G_\epsilon \equiv U[\bar{\xi} - \epsilon, \bar{\xi} + \epsilon]$. The induced cost function is

$$C_\epsilon(\varphi) = w(\bar{\xi} - \epsilon)\varphi + w\epsilon\varphi^2, \quad (5)$$

which is strictly convex with $C_\epsilon'''(\varphi) = 2w\epsilon > 0$ for every $\epsilon > 0$. As $\epsilon \rightarrow 0$:

- (i) $C_\epsilon(\varphi) \rightarrow w\bar{\xi}\varphi$ pointwise: the linear (degenerate) cost.
- (ii) $C_\epsilon'''(\varphi) \rightarrow 0$: curvature vanishes.
- (iii) The optimal $\varphi_\epsilon^*(S) = \max\left\{0, \min\left\{\frac{S - w(\bar{\xi} - \epsilon)}{2w\epsilon}, 1\right\}\right\}$ converges to $\mathbf{1}[S \geq w\bar{\xi}]$: the binary threshold rule.

Thus the deterministic fixed cost is the singular limit of a family of strictly convex problems.

Proof. For $G_\epsilon = U[\bar{\xi} - \epsilon, \bar{\xi} + \epsilon]$, applying (3) gives $C_\epsilon(\varphi) = w(\bar{\xi} - \epsilon)\varphi + w\epsilon\varphi^2$, with $C''_\epsilon = 2w\epsilon > 0$. The full derivation is in Online Appendix A. ■

The result recasts the standard view that convex costs are a “smooth approximation” to the “true” lumpy friction. The deterministic fixed cost is the knife-edge limit; arbitrarily close perturbations restore strict convexity. The curvature $C''_\epsilon = 2w\epsilon$ measures the “softness” of the adjustment margin: even an infinitesimally small ϵ produces a strictly convex problem.

Epistemic interpretation and continuity in G The deterministic fixed cost requires the agent to know the adjustment cost with infinite precision — an idealisation rarely satisfied in practice (staff time, managerial attention, installation logistics) — whereas any ϵ -band of noise makes the decision problem smooth and convex. The deterministic case is fragile under perturbation; the adjustment-probability representation varies continuously under standard regularity on G (g continuous and bounded away from zero on the relevant range). The set of distributions generating strictly convex C is dense under the weak topology: every neighbourhood of a point mass contains absolutely continuous distributions with strictly convex induced costs (Online Appendix A).

2.5 Dispersion and the cost of adjustment

Proposition 2 establishes that the point mass is the singular case. This subsection shows it is also the most expensive one: holding the mean cost fixed, the concentration of G — not the level of costs — determines how expensive adjustment is. The result is a comparative static in dispersion with direct economic content.

Proposition 3 (Dispersion and convexity).

Let G_1 and G_2 be distributions on $[0, \bar{\xi}]$ with the same mean and let G_2 be a mean-preserving spread of G_1 . The induced costs satisfy:

- (i) $C_1(1) = C_2(1) = w E[\xi]$.
- (ii) $C_2(\varphi) \leq C_1(\varphi)$ for all $\varphi \in [0, 1]$: greater dispersion lowers the cost function in the interior, via the integrated quantile order $\int_0^t G_2^{-1} \leq \int_0^t G_1^{-1}$ characterising MPS.
- (iii) Within the power law family, the mean-matched marginal cost schedules $C'_\gamma(\varphi) = w\bar{\xi}_\gamma \varphi^{1/\gamma}$, with $\bar{\xi}_\gamma = (1 + 1/\gamma) E[\xi]$ holding the mean fixed across γ , cross exactly once in the interior $(0, 1)$ as γ varies (pairwise single-crossing; the crossing point is

normalisation-dependent). Outside parametric families, multiple crossings of $C'_2 - C'_1$ are possible without an additional single-crossing assumption.

The degenerate distribution maximises C on $(0, 1)$ and is the unique case in which C is linear; any non-degenerate G with positive density makes C strictly convex.

Proof. See Online Appendix A. ■

The mechanism is the option value of selection under a mean-preserving spread. The spread relocates probability mass from the centre of G toward both tails, leaving the mean (and with it the full-adjustment cost $C(1) = w E[\xi]$) unchanged. But the threshold rule does not draw on the two tails symmetrically: adjusters are recruited from the bottom of the cost distribution upward, so at any interior adjustment rate φ the agent exploits the heavier low-cost tail while the heavier high-cost tail is left entirely to the non-adjusters. The benefit of the extra cheap draws is collected; the burden of the extra expensive draws (the other half of the mean-preserving trade) is never paid. This is why $C(\varphi)$ falls everywhere in the interior even though the average cost is held fixed. The degenerate distribution is the polar opposite — every adjuster pays the full mean cost, with no cheap tail to exploit — and is therefore the *most expensive* configuration of a given average cost, not a neutral benchmark. Heterogeneity in adjustment costs is not a nuisance to be averaged out; it is itself a reduction in the effective friction.

The macro reading is a caution about measurement and calibration. Two economies with the same measured average adjustment cost can face very different effective frictions, so mean cost measurements overstate the friction whenever costs are dispersed. [Gourio and Kashyap \(2007\)](#) make the quantitative version of this point in lumpy investment: the aggregate implications of fixed costs hinge on the calibrated cost *distribution*, not just its mean. Proposition 3 gives that sensitivity an analytical form — an MPS of G shifts the entire cost function C , with the mean cost held fixed. The same comparative static reappears in the pricing application: within the power law family, γ indexes concentration, and the concentration that raises the effective cost of adjustment is precisely what strengthens selection and steepens the Phillips curve (Section 3). Because the realised welfare cost of the friction is $C(\varphi^*)$ (Section 2.7), the dispersion of G directly lowers the welfare burden of adjustment at any given adjustment rate.

2.6 Fixed costs as generalised entropies

This subsection develops the C -side correspondence: for researchers whose primitive is a convex penalty C on choice probabilities, the duality reveals which underlying distribution

G they have implicitly assumed. The cost function $C(\varphi)$ is fully determined by G and inherits its shape from the quantile function G^{-1} . Canonical distributions G yield cost functions that coincide with well-known information-theoretic entropies. The implication is that the choice of cost specification in any model with frictional adjustment is, implicitly, an assumption on the underlying distribution G of fixed costs.

Proposition 4 (Entropy correspondence).

The closed forms of Corollary 2 have known information-theoretic counterparts:

- (i) **Uniform distribution:** the quadratic cost of Corollary 2(i) is the L^2 penalty of regularisation frameworks.
- (ii) **Power law distribution:** the iso-elastic cost of Corollary 2(ii), proportional to φ^q with $q = 1 + 1/\gamma$, shares its exponent with the Tsallis q -divergence from the no-adjustment prior (Tsallis, 1988).
- (iii) **Exponential distribution:** the cost of Corollary 2(iii) is the mutual-information cost of binary rational inattention (Sims, 2003; Matějka and McKay, 2015). Formally it is the generalised Kullback-Leibler (I-)divergence of the adjustment probability from the no-adjustment default, the Bregman divergence generated by $h(x) = x \ln x$.²

Proof. See Online Appendix A. ■

Two boundaries of the correspondence are worth marking. Within the power law family, γ controls the exponent ($\gamma = 1$ gives $q = 2$, quadratic; $\gamma \rightarrow \infty$ gives $q \rightarrow 1$, linear), but the logarithmic (Shannon) form is recovered only by the exponential distribution, not by any power law limit. And the adjustment probability under exponential costs, $\varphi^* = 1 - e^{-\lambda S/w}$, is the exponential CDF evaluated at S/w — *not* the logistic function $1/(1 + e^{-S/\sigma})$ of logit discrete choice; the two share a sigmoid shape but differ in functional form.

The correspondences are of functional form, not of economic mechanism: uniform G yields the quadratic attention penalty of Gabaix (2014); exponential G yields the KL divergence used in binary rational inattention (Sims, 2003; Matějka and McKay, 2015); in the multi-action extension (Section 2.9), i.i.d. Gumbel costs yield negative Shannon entropy and the multinomial logit. The reduced-form $C(\varphi)$ is the same; welfare and identification implications, taken up in Section 2.7, differ.

²Strict Shannon KL from a degenerate prior is infinite for any positive φ ; the cost here is the generalised KL / I-divergence, finite on the simplex interior.

The distribution G as a Rosetta Stone Table 1 summarises the mapping. Five literatures — menu costs, lumpy investment, discrete choice, rational inattention, and control costs — have independently developed cost specifications that are the parametric image of an implicit assumption on G . Making this assumption explicit converts a free modelling choice into a parametric restriction on a cost distribution that can be measured.

Table 1: The distribution G as a Rosetta Stone

Distribution G	Cost $C(\varphi)$	Form	Entropy	Literature
Uniform ($\gamma=1$)	φ^2	Quadratic	L^2	Sparsity
Power law (γ)	$\varphi^{1+1/\gamma}$	Iso-elastic	Tsallis- q exp. only	Generalised discrete choice
Exponential	$(1-\varphi)\ln(1-\varphi) + \varphi$	Log. MC	KL divergence	Rational inattention
Degenerate	$\xi \varphi$ on $\{0, 1\}$	Linear	—	Lumpy adjustment
Type I EV (multinomial)	$\sum \varphi_j \ln \varphi_j$	Neg. entropy	Shannon (mv)	Logit discrete choice

Notes: Each row specifies a distribution G of fixed costs. The second column gives the induced convex cost function $C(\varphi)$ via equation (3) (up to scale). The fourth column identifies the information-theoretic entropy to which C corresponds. The last column indicates the literature where this cost function appears. “Log. MC” = logarithmic marginal cost. “mv” = multivariate. The first four rows are binary choice; the last row is the multi-action extension (Proposition 5). The exponential→KL row is an exact functional identity, whereas the power-law row shares only the exponent $q = 1 + 1/\gamma$ of the Tsallis q -divergence, not the full two-argument functional. Reconciling the five literatures with the rows: menu costs are the generic fixed-cost reading specialised by every row; lumpy investment is the degenerate (lumpy-adjustment) row; rational inattention the exponential; control costs the uniform (sparsity) penalty; and discrete choice appears twice, as the iso-elastic generalised case and the logit limit. The Type I EV row is a signed payoff-shock primitive (taste shocks, isomorphic under a sign change), not a non-negative cost distribution.

2.7 Welfare content of the duality

The welfare cost of adjustment at a given equilibrium is

$$\text{Welfare cost} = w \int_0^{G^{-1}(\varphi^*)} \xi dG(\xi) = C(\varphi^*), \quad (6)$$

a closed-form measure for fixed-cost models, with $\varphi^* = G(S/w)$ from the smooth FOC $S = C'(\varphi)$ (and φ^* still set by the full equilibrium). The reverse reading gives information costs a physical interpretation: in rational inattention the mutual-information cost $\lambda \sum \varphi_j \ln \varphi_j$ is measured in nats with λ an abstract shadow price, but under the binary exponential- G primitive $C(\varphi^*)$ is the expected physical cost paid by adjusters and λ is the rate parameter of G , in principle estimable from micro data. This resource-cost reading requires non-negative physical-cost primitives; the Gumbel/multi-action logit case (Proposition 5(ii)) is a reduced-form conjugacy without it.

2.8 Theoretical and computational implications

This subsection develops the G -side payoff: for a researcher whose primitive is a cost distribution G , the duality delivers a smooth dynamic value function, a closed-form first-order condition, and direct integration into FOC-based solvers. The duality applies state by state in a Bellman with extensive choice $p \in [0, 1]$ and intensive target x' .

Theorem 2 (Differentiability of the lumpy adjustment value function).

Consider the dynamic problem of Section 3.1 in adjustment / non-adjustment branch form. Assume G is absolutely continuous with density $g \in C^1$, $g > 0$ on the interior of its support; the per-period payoffs are bounded and C^1 , with F^a moreover C^2 and strictly concave in the intensive target x' ; and, on compact state spaces, the resetting firm's full objective is uniformly strictly concave in x' with a non-degenerate second-order condition — the joint compactness / interior-optimum / strict-SOC hypothesis of Santos (1991), Theorem 4.1, stated on the full objective rather than the per-period payoff alone (precise conditions in Online Appendix G). Then the Bellman operator has a unique fixed point $v(\cdot; X)$ in bounded measurable functions, C^1 in x for each X , with v_x given by the Benveniste-Scheinkman envelope.

Proof sketch. By (A1) and Proposition 1, the Fenchel conjugate $c^*(S; X) = \max_{p \in [0, 1]} \{pS - C(p; X)\}$ is C^1 in S , so the Bellman concentrates to $v(x; X) = v^n(x; X) + c^*(S(x; X); X)$. A unique bounded fixed point exists by Blackwell's conditions; the operator preserves bounded C^1 functions because the resetting firm's full maximand satisfies the compactness, interior-optimum, and uniform-strict-concavity hypotheses of Santos (1991), Theorem 4.1, under (A2)–(A3), delivering a C^1 value with the Benveniste-Scheinkman envelope (Benveniste and Scheinkman, 1979). Here (A3), uniform strict concavity of the resetting firm's full maximand, is a sufficient regularity condition on the objective (the iterate), not a primitive restriction on G ; it is stated as a curvature-dominance inequality and verified across the spectrum in Online Appendix G. The full argument is in Online Appendix G.

The contribution of the duality at this step is to make the smoothness of the ex-ante value operational. The C^1 regularity itself is standard, following from general dynamic-programming arguments (Santos 1991) and obtained in lumpy settings via smooth-pasting at inaction-region boundaries (Baley and Blanco 2021). The cost object coincides with that of Alvarez, Lippi, and Oskolkov (2022), recovered there from the hazard by solving a second-order linear ordinary differential equation in the value function. *What is new here is the route:* the map from the cost distribution to the convex cost, $G \mapsto C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi)$, is environment-free, depending only on (G, w) with the surplus S left abstract, and it yields a closed-form first-order condition $S = C'(\varphi)$ in any environment

admitting a scalar adjustment surplus. Methods that work with the realised threshold rule (or the Ss-band cutoff in x for given ξ) face a kink in the conditional value function and require threshold-specific machinery; smooth-pasting approaches handle this at the inaction-region boundaries through optimal-control conditions; ODE-based approaches construct v from a Sturm-Liouville-type solution. The Fenchel-dual reformulation supplies a smooth maximand globally (the convex cost C derived from the absolutely continuous G via Proposition 1) to which Santos’ theorem applies directly. This bridge, not the C^1 conclusion, is what Theorem 2 contributes. ■

Corollary 3 (Policy differentiability).

Under (A1)–(A3), the intensive policy $x^(x; X)$ is C^1 in x for each X , and the extensive policy $p^*(x; X) \in [0, 1]$ is continuous on (x, X) and C^1 on the interior set $\{(x, X) : p^*(x; X) \in (0, 1)\}$.*

Proof. The two margins are treated separately (the joint objective is not jointly concave). The intensive policy x^* is C^1 by the implicit function theorem on its strictly concave FOC; the extensive policy solves $C_p(p^*; X) = S(x; X)$ and is C^1 on the interior of $\{p^* \in (0, 1)\}$. See Online Appendix G. ■

Because the ex-ante value function and the policies are smooth in the state, standard equilibrium-existence and comparative-statics arguments apply directly, without the threshold-specific machinery that a kinked conditional value function would require.

Computational routine With $v \in C^1$ and differentiable policies, the routine runs in closed FOC–envelope form: the extensive margin solves $S = C'(p)$ (with a KKT projection at the boundary) and the intensive target x^* comes from the FOC–envelope system without grid search, in the endogenous-grid style of Carroll (2006). This slots the model into FOC-based heterogeneous-agent toolkits (Reiter, 2009; Auclert, Rognlie, and Straub, 2020) and supports the global non-linear solution of Section 4.5. The full routine is in Online Appendix G.

2.9 Beyond binary choice

The duality extends to multi-action settings via the same conjugate construction. Suppose the agent chooses among $\{a_0, a_1, \dots, a_J\}$ (a_0 = inaction), with a_j ($j \geq 1$) yielding surplus S_j at random cost $w\xi_j$, $\xi_j \sim G_j$ independent across actions, and selects the action with the highest net payoff.

Proposition 5 (Multi-action duality).

Let $J \geq 1$ and suppose $\xi_j \sim G_j$ independently for $j = 1, \dots, J$, with each G_j absolutely continuous with positive density on $[0, \bar{\xi}_j]$. Define $S_0 = 0$, $\xi_0 = 0$. Then:

(i) The expected payoff $\Pi^{FC}(\mathbf{S}) = E[\max_{j \in \{0, \dots, J\}} \{S_j - w\xi_j\}]$ equals

$$\Pi^{PC}(\mathbf{S}) = \max_{\boldsymbol{\varphi} \in \Delta^J} \sum_{j=0}^J \varphi_j S_j - C(\boldsymbol{\varphi}), \quad (7)$$

where $C : \Delta^J \rightarrow \mathbb{R}$ is jointly convex and determined by the joint distribution of $(\xi_1, \dots, \xi_J)$, which under independence is pinned down by the marginals $(G_1, \dots, G_J)$.

(ii) (Payoff-shock analogue.) Reinterpret ξ as a real-valued payoff shock rather than a non-negative physical cost (so $\eta_j \equiv -w\xi_j$ plays the role of a Gumbel taste shock in the standard random-utility setup). When $\eta_j \stackrel{i.i.d.}{\sim} \text{Gumbel}(0, \sigma)$ for all $j \geq 0$, the conjugate construction yields $C(\boldsymbol{\varphi}) = \sigma \sum_{j=0}^J \varphi_j \ln \varphi_j \leq 0$ (negative Shannon entropy on Δ^J). The optimal choice probabilities are

$$\varphi_j^* = \frac{e^{S_j/\sigma}}{\sum_{k=0}^J e^{S_k/\sigma}}, \quad (8)$$

the multinomial logit of [McFadden \(1974\)](#).

Proof.

Apply the Fenchel conjugate construction to the multi-alternative maximum and specialise to the Gumbel distribution. The detailed proof is given in Online Appendix A. ■

Form of the multi-action C In the binary case $C(\varphi) = w \int_0^\varphi G^{-1}(u) du$. The multi-action cost has no such explicit form in general: it is the Fenchel conjugate of $\Pi^{FC}(\mathbf{S})$,

$$C(\boldsymbol{\varphi}) = \sup_{\mathbf{S} \in \mathbb{R}^{J+1}} \left\{ \sum_{j=0}^J \varphi_j S_j - \Pi^{FC}(\mathbf{S}) \right\},$$

with marginal-cost interpretation $\partial C / \partial \varphi_j = S_j$ on the simplex interior. Explicit closed forms exist when the joint distribution of $(\xi_0, \dots, \xi_J)$ admits one: i.i.d. Gumbel costs give the negative Shannon entropy of part (ii), and the binary case reduces to the quantile integral of Theorem 1. The reason the binary case is special is that $\boldsymbol{\varphi}$ collapses to a one-dimensional parameter, and the cost paid by the marginal adjuster is the cost-quantile of the action's threshold. With multiple alternatives, the conditional cost paid given that action j is chosen

depends on order-statistic comparisons across all alternatives, so the simple quantile-of-the-threshold reading does not carry through.

When the surplus depends on a state x , the duality applies state by state: $\varphi^*(x) = G(S(x)/w)$ is the endogenous state-dependent hazard. In dynamic settings the probability-choice formulation gives an explicit smooth first-order condition for the resetter; implementations that work with the realised threshold rule face a kink in the conditional value function at the adjustment threshold, and the degenerate- G (Golosov-Lucas) limit produces a corner kink even in the ex-ante value.

2.10 Application to investment

The duality has a direct implication for the lumpy investment literature. In the [Thomas \(2002\)](#) model, [Gourio and Kashyap \(2007\)](#) show that the shape of the fixed cost distribution is the key determinant of whether lumpy micro-level investment spikes matter for aggregates. The duality makes this precise: each G maps to a unique convex cost C , and the shape of G governs the curvature of C and hence the aggregate investment elasticity. Under uniform G , C is quadratic ([Corollary 2](#)), so each firm’s investment problem is individually convex — providing an independent partial-equilibrium smoothness channel, distinct from the general-equilibrium aggregation debate of [Khan and Thomas \(2008\)](#). The GE channel (endogenous factor price adjustment dampening investment spikes) provides another route to smoothness; the PE convexity established here provides an independent sufficient condition that does not rely on GE feedbacks. More broadly, any G with a positive density delivers smoothness at the individual level. The dispersion of G provides a continuous index of lumpiness, with the two ends of the spectrum tracing out distinct adjustment patterns. Concentrated G (high γ) means costs cluster near $\bar{\xi}$ (similar across firms), so adjustment occurs in sharp spikes when surpluses cross the common threshold. Dispersed G (low γ), by contrast, spreads costs widely, producing smoother adjustment. As dispersion shrinks to a point mass, the model recovers the binary threshold rule. [Oskolkov and Lippi \(2026\)](#) recover G from plant-level investment data, confirming that the shape of the cost distribution varies across sectors. Formal conditions for smoothness and the investment elasticity formula are in [Online Appendix B](#).

3 The menu cost–Calvo spectrum

The deepest application of the duality is to price setting, where the distinction between time-dependent (Calvo) and state-dependent (menu cost) pricing is central to New Keyne-

sian economics. The idea that these models sit on a spectrum is not new: [Dotsey, King, and Wolman \(1999\)](#) draw menu costs from a distribution, and [Caballero and Engel \(1993\)](#) parametrise the adjustment hazard directly. What the duality plus the power-law parametrisation add is a closed-form analytical decomposition: the repricing hazard $\Lambda(x)$ is derived from G rather than assumed, and the frequency-selection decomposition is indexed by a single parameter. The distribution G governs the repricing hazard, the selection effect, the Phillips curve slope, and the degree of monetary non-neutrality.

This section follows an analytical strategy: replace the dynamic repricing surplus by its quadratic approximation $S(x) = Bx^2$ and derive everything (the hazard, the Phillips curve slope $\kappa = (1 + 2\gamma)\bar{\Lambda}$, the decomposition into frequency and selection) in closed form. The choice is for clarity, not necessity, since each result can also be obtained without the approximation. The full dynamic model (9)–(14) is itself solvable globally, and [Section 4.5](#) does so. The reason to lead with the analytical approach is that closed-form expressions show which primitive moves which aggregate margin, a mapping that survives recalibration. A numerical solution at a single point of the parameter space cannot deliver this mapping on its own. The global solution then plays a complementary role: it checks numerically how well the closed-form decomposition approximates the unapproximated dynamic model. The dynamic surplus is only approximately quadratic, so $\kappa = (1 + 2\gamma)\bar{\Lambda}$ holds to that approximation, not as an exact identity.

3.1 The dynamic pricing problem

A note on notation is needed first, because a single object travels under three names across the paper. The generic extensive-margin choice probability that I called φ in [Section 2](#) is renamed Λ here to match menu-cost convention (the per-period repricing probability and, equivalently, the firm-level hazard); the generic Bellman in [Theorem 2](#) and [Online Appendix G](#) uses p for the same object. All three letters refer to the same extensive-margin choice probability.

The application embeds the duality in a dynamic New Keynesian environment. I summarise the firm block here; the household block, the law of motion for the price-gap distribution, the price aggregator, and monetary policy are the standard New Keynesian closure and are stated in full in [Online Appendix G](#). A monopolistically competitive firm has current log price p and desired log price p^* , with *price gap* $x \equiv p - p^*$. A non-adjusting firm’s gap drifts by $-\pi'$ each period through aggregate inflation, which raises the desired price while the posted price stands still; an adjusting firm resets it to the optimal target.³ Let X denote the

³Idiosyncratic shocks to the desired price can be added. They superimpose diffusion on the gap process and shape the cross-section, but leave the duality, the hazard $\Lambda(x) = G(Bx^2/w)$, and the frequency-selection

aggregate state (the cross-sectional gap distribution and the exogenous shocks), $R(x; X)$ the per-period real profit at gap x , β the discount factor, and $m(X, X')$ the stochastic discount factor.

Each period the firm chooses a repricing probability $\Lambda \in [0, 1]$, paying the convex cost $C(\Lambda)$ that the duality (Theorem 1) assigns to the menu-cost distribution G . Its value function solves

$$V(x; X) = \max_{\Lambda \in [0, 1]} \left\{ \Lambda V^a(X) + (1 - \Lambda) V^n(x; X) - C(\Lambda) \right\}, \quad (9)$$

$$V^a(X) = \max_{x'} \left\{ R(x'; X) + \beta \mathbb{E} \left[m(X, X') V(x' - \pi'; X') \right] \right\}, \quad (10)$$

$$V^n(x; X) = R(x; X) + \beta \mathbb{E} \left[m(X, X') V(x - \pi'; X') \right], \quad (11)$$

where $V^a(X)$ is the value of repricing (resetting the gap to the optimal target x'^*) and $V^n(x; X)$ the value of not repricing. The *repricing surplus* is

$$S(x; X) \equiv V^a(X) - V^n(x; X), \quad (12)$$

and the duality's first-order condition $S = C'(\Lambda)$ delivers the state-dependent hazard in closed form,

$$\Lambda(x; X) = G\left(\frac{S(x; X)}{w}\right) = \min\left\{\left(\frac{S(x; X)}{w\xi}\right)^\gamma, 1\right\}, \quad (13)$$

the second equality holding for the power law $G(\xi) = (\xi/\bar{\xi})^\gamma$. By Theorem 2 and Corollary 3, V is C^1 in x , so the reset target solves the smooth first-order condition

$$\begin{aligned} R_x(x'^*; X) + \beta \mathbb{E} \left[m(X, X') V_x(x'^* - \pi'; X') \right] &= 0, \\ V_x(x; X) &= (1 - \Lambda(x; X)) V_x^n(x; X), \end{aligned} \quad (14)$$

the second equation being the standard envelope condition (Benveniste and Scheinkman, 1979). There is no threshold to locate and no firm-level grid search: the entire policy reduces to a smooth first-order condition, which is what lets the model slot into standard heterogeneous-agent solvers. The dynamic Bellman (9)–(14) above is used in two places in what follows. The next subsection replaces the dynamic surplus by its quadratic approximation and derives the closed-form analytical results of this section from there. Section 4.5

decomposition unchanged. Crucially, the differentiability of the value function (Theorem 2) comes from the convexity of the Fenchel-dual cost C , not from idiosyncratic smoothing of the value function as in the menu-cost tradition. The cross-section is therefore kept non-degenerate by trend inflation alone, as in the global computation of Section 4.5 (with implementation details in Online Appendix G). Including idiosyncratic shocks does not change the results.

returns to the full dynamic system and solves it globally, both as a check on the analytical approximation and to trace how the tail of G shapes dynamic transmission, which a closed-form approach cannot.

3.2 Static approximation: the repricing hazard, cross-section, and selection

For the analytical results that follow I specialise the repricing surplus (12) to its second-order approximation around the optimum. To second order the gain from closing the gap is quadratic, and I adopt this static form for analytical clarity:

$$S(x) = Bx^2, \tag{15}$$

where $B > 0$ depends on the demand elasticity and the discount factor.⁴ The loss from mispricing grows with the square of the gap, and under this quadratic specialisation and the interior-support condition below, the static Fenchel mapping is exact.

The firm draws a menu cost $\xi \sim G$ each period and reprices if and only if the gain from repricing exceeds the cost $w\xi$, where $w > 0$ is the wage (as in Section 2):

$$\text{Reprice if and only if } w\xi \leq Bx^2. \tag{16}$$

What the dynamic surplus would change, and what it would not In the full dynamic problem the surplus is $S(x; X) = V^a(X) - V^n(x; X)$, which differs from the static Bx^2 by an option-value term and a larger curvature (it accumulates discounted future flow losses). It remains approximately quadratic near the optimum, which is all the aggregate decomposition needs; and since $\kappa/\bar{\Lambda} = 1 + 2\gamma$ is independent of the level of B (Online Appendix C), the same closed form applies with the dynamic B (Alvarez, Le Bihan, and Lippi, 2016). The duality (Theorem 1) holds for any S ; the quadratic is for clarity.

The repricing hazard By Theorem 1, the firm's repricing decision under the fixed cost formulation (Definition 1) is equivalent to the probability choice formulation (Definition 2) with convex cost $C(\varphi)$. The first-order condition $Bx^2 = C'(\varphi) = wG^{-1}(\varphi)$ yields the optimal repricing probability as a function of the price gap:

⁴This is standard; see Golosov and Lucas (2007) and Alvarez, Le Bihan, and Lippi (2016). The quadratic loss approximation is $S(x) \approx \frac{\varepsilon}{2}x^2$ with ε the demand elasticity, up to discounting.

Definition 3 (Repricing hazard).

The repricing hazard, the probability that a firm with price gap x adjusts its price, is

$$\Lambda(x) \equiv G\left(\frac{Bx^2}{w}\right). \quad (17)$$

This is the central object of the pricing application: not a numerical output but an explicit formula from the duality and the choice of G .⁵ Conditional on (B, w) and f_{ss} , all aggregate pricing implications flow through Λ , so G is the structural primitive.

Proposition 6 (Hazard properties).

The repricing hazard $\Lambda(x)$ satisfies:

- (i) $\Lambda(0) = G(0) = 0$: a firm at its optimal price never reprices. The gain from doing so is zero, and every cost draw is positive.
- (ii) Λ is symmetric in x and increasing in $|x|$: the repricing probability depends only on the magnitude of the gap, not its sign. The surplus Bx^2 depends only on x^2 . Larger gaps produce larger surpluses, which exceed more cost draws.
- (iii) $\Lambda'(x) = g(Bx^2/w) \cdot 2Bx/w$: the sensitivity depends on the density g evaluated at the firm's surplus. Where g is high, many firms are near the threshold, and a small change in the gap tips many across it. Where g is low, few firms are near the threshold, and the same change has little effect.
- (iv) If G has bounded support $[0, \bar{\xi}]$, then $\Lambda(x) = 1$ for $|x| \geq \sqrt{w\bar{\xi}/B}$: when the gap is sufficiently large, the surplus exceeds even the highest cost draw, and the firm reprices with certainty. This defines the “inaction band” beyond which all firms adjust. If G has unbounded support (e.g., exponential), $\Lambda(x) < 1$ for all finite x and $\Lambda(x) \rightarrow 1$ only as $|x| \rightarrow \infty$.

How quickly Λ rises from zero to one is governed by the density g at the relevant surplus level: the mechanism through which G shapes aggregate dynamics.

Microfoundation of the hazard elasticity The hazard $\Lambda(x) = G(S(x)/w)$ has a well-defined log-log local elasticity in $|x|$, decomposable by the chain rule into two structural

⁵ $\Lambda(x)$ is the “adjustment function” of Caballero and Engel (1993, 1999), taken there as a primitive; here it is derived from G , the deeper structural object.

factors:

$$\nu \equiv \frac{d \log \Lambda(x)}{d \log |x|} = \underbrace{\frac{d \log S(x)}{d \log |x|}}_{\text{surplus elasticity}} \cdot \underbrace{\left. \frac{d \log G(z)}{d \log z} \right|_{z=S(x)/w}}_{\text{cost-shape exponent}}. \quad (18)$$

For the power-law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ the second factor is γ ; for the quadratic surplus the first is 2, so

$$\nu = \frac{d \log S}{d \log |x|} \cdot \gamma = 2 \cdot \gamma = 2\gamma, \quad (19)$$

with $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$. The exponent thus has a structural anatomy (ν is surplus curvature times cost concentration) and labels the horizontal axis of the two-dimensional map of Section 3.4: the local exponent $\nu = 2\gamma$ that Alvarez, Lippi, and Oskolkov (2022) recover identifies one coordinate of G , the cost-concentration parameter, leaving the global tail free.

Twin identity of γ : G -side and C -side readings The duality’s sharpest empirical consequence is that the cost-concentration parameter γ admits two simultaneous structural readings. On the G -side, γ is the local exponent of the menu-cost distribution and indexes the entire Calvo–Golosov–Lucas spectrum, with hazard exponent $\nu = 2\gamma$ and Phillips slope $\kappa = (1 + 2\gamma)\bar{\Lambda}$. On the C -side, the dual cost is iso-elastic in the adjustment probability, $C(\varphi) = \frac{\gamma}{\gamma+1} w \bar{\xi} \varphi^{1+1/\gamma}$, with elasticity $\alpha = 1 + 1/\gamma$. Iso-elastic costs of this form are the generalised-entropy / α -divergence control costs of the entropy correspondence (Proposition 4; the Shannon point is exponential G). The *same* empirical anchor identifies both readings: ALO’s recovered $\nu \approx 2$ implies $\hat{\gamma} = 1$, which is simultaneously the Khan-Thomas uniform- G point on the menu-cost side and a quadratic ($\alpha = 2$) attention cost on the information-cost side. The duality is what makes the calibration shared: a single number, drawn from price-change microdata, structurally identifies parameters of two cost specifications that the literature has previously treated as unrelated.

The tail is not locally identified The exponent $\nu = 2\gamma$ is a *local* property of the hazard at small gaps: any G with positive density at the origin gives $\nu = 2$, so the local moment identifies the cost-concentration coordinate of G but leaves the global tail free. Uniform G (Khan-Thomas), exponential G (Shannon RI), and heavy-tail Lomax/Pareto G all satisfy $\nu \approx 2$ yet differ in their right tails. Section 4.5 solves the dynamic model across this tail dimension and finds that the tail, invisible to the local moment, shapes the selection response, and with it transmission (Figure 4). Disambiguating within the class therefore

requires moments outside the local-hazard family, of which the right tail of the price-change distribution is the most natural.

The stationary distribution of price gaps To close the model, the cross-sectional gap density f_{ss} must be characterised. Under trend inflation $\pi > 0$ a non-adjusting firm's gap falls by π each period while adjusters reset to x^* ; the cross-section is renewed by resets and stretched by drift.

Proposition 7 (Stationary distribution).

Under $\Lambda(x) = G(Bx^2/w)$ and trend inflation $\pi > 0$, the stationary gap density satisfies the renewal fixed point

$$f_{ss}(x) = \underbrace{\bar{\Lambda} r(x)}_{\text{reset flow}} + \underbrace{(1 - \Lambda(x + \pi)) f_{ss}(x + \pi)}_{\text{drift flow}}, \quad \bar{\Lambda} = \int \Lambda(x) f_{ss}(x) dx, \quad (20)$$

where r is the post-reset gap distribution: degenerate at the reset point absent idiosyncratic shocks, smoothed by them otherwise. Equivalently, $f_{ss}(x)$ is proportional to the probability that a firm has not repriced since its last reset, so it decays along the drift direction at a rate governed by the hazard. Under positive trend inflation f_{ss} is asymmetric, supported on the inaction region between the reset point and the adjustment boundary. The proof is in Online Appendix C.

A concentrated G (steep hazard) yields a tight distribution near the reset point; a dispersed G (flat hazard) lets firms drift far, yielding a wide one.

The selection effect Under state-dependent pricing, firms with the largest gaps are most likely to reprice, amplifying the aggregate response relative to Calvo (Goloso and Lucas, 2007). The duality characterises how G governs this selection: I define a selection-intensity measure, then link it to G .

Definition 4 (Selection intensity).

Given a cross-sectional density $f(x)$ of price gaps, the average absolute adjusted price change is defined as

$$\overline{|x|}^{adj} := \frac{\int |x| \cdot \Lambda(x) \cdot f(x) dx}{\int \Lambda(x) \cdot f(x) dx} = \frac{E[\Lambda(x) \cdot |x|]}{E[\Lambda(x)]}. \quad (21)$$

This is the mean absolute price change among repricing firms. The selection intensity is the proportional excess of the average adjuster's absolute price change over the population

average:

$$\mathcal{S} \equiv \frac{\overline{|x|}^{adj}}{E[|x|]} - 1. \quad (22)$$

When Λ is constant (Calvo), $\mathcal{S} = 0$; when Λ is a step function (Goloso-Lucas), $\mathcal{S}$ attains its polar value within the power-law family at fixed $\bar{\Lambda}$. Since $\Lambda(x) = G(Bx^2/w)$, the shape of G sets selection strength.

Proposition 8 (Selection and the menu cost distribution).

Fix the cross-sectional density f . The adjustment-weighted mean $\overline{|x|}^{adj}$ admits the decomposition

$$\overline{|x|}^{adj} = E[|x|] + \frac{\text{Cov}(\Lambda(x), |x|)}{\bar{\Lambda}},$$

so that selection intensity $\mathcal{S}$ has the same sign as $\text{Cov}(\Lambda(x), |x|)$. As a consequence:

- (i) $\mathcal{S} = 0$ in the Calvo limit (constant Λ), since the covariance vanishes.
- (ii) $\mathcal{S} > 0$ whenever Λ is strictly increasing in $|x|$. The size of $\mathcal{S}$ scales with the local elasticity $\partial \ln \Lambda / \partial \ln |x| = 2(g/G)(Bx^2/w) \cdot (Bx^2/w)$, i.e., twice the reversed hazard rate of G at the firm's surplus.
- (iii) In the Goloso-Lucas limit, Λ is a step function and adjusters are concentrated at the inaction-band boundary; this is the polar case of the spectrum, with $\mathcal{S}$ the largest attained within the power law family at fixed $\bar{\Lambda}$. A general extremality statement across all hazards calibrated to the same average frequency would require a rearrangement-style argument not pursued here.

Proof. See Online Appendix C. ■

A concentrated G gives a steep hazard and strong selection; a dispersed G gives a flat hazard and weak selection. The density g at the threshold sets how many firms a small gap increase tips into adjustment.

3.3 The Calvo–Goloso-Lucas spectrum and the two-dimensional map

The results so far hold for any G with positive density, so to organise them I now restrict attention to a tractable one-parameter family that connects the two canonical models, Calvo and Goloso-Lucas, as polar shapes of the hazard.

Calvo (1983) In the Calvo (1983) model, each firm reprices with a fixed probability $\lambda \in (0, 1)$ regardless of its price gap. Since $\Lambda(x) = G(Bx^2/w)$ is always increasing in $|x|$, a strictly constant hazard cannot arise from any non-degenerate absolutely continuous G . The framework therefore does not literally nest Calvo; the following proposition makes the limiting sense precise.

Proposition 9 (Calvo approximation).

For the power law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ with $\bar{\xi}(\gamma)$ calibrated to maintain a fixed average frequency λ , and assuming f_{ss} is supported in the interior of the inaction band, the selection wedge $\kappa - \lambda = 2\gamma\lambda$ vanishes as $\gamma \rightarrow 0^+$. In this limit:

- (i) *The Phillips curve slope converges to the Calvo slope: $\kappa \rightarrow \lambda$.*
- (ii) *Non-neutrality converges to the Calvo level: $1 - \kappa \rightarrow 1 - \lambda$.*

The hazard $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$ does not converge to a constant as $\gamma \rightarrow 0^+$ with fixed $\bar{\xi}$: it converges to near-certain adjustment for any $x \neq 0$ (since $a^\gamma \rightarrow 1$ for any $a > 0$). The Calvo approximation arises only through the frequency-calibrated limit, in which $\bar{\xi} \rightarrow \infty$ as $\gamma \rightarrow 0^+$ so that the hazard remains small and approximately constant over the support of f_{ss} (the residual variation goes as $\gamma \log(Bx^2/(w\bar{\xi}))$, vanishing in the limit). In this regime selection contributes $2\gamma\lambda$ to the Phillips curve slope, vanishing relative to the frequency component λ .

Proof. See Online Appendix C. ■

Golosov and Lucas (2007): step-function hazard When G is degenerate at $\bar{\xi}$, $\Lambda(x) = \mathbf{1}[Bx^2 \geq w\bar{\xi}]$, the canonical Ss rule with inaction band $\bar{x} = \sqrt{w\bar{\xi}/B}$. This is the polar case of the spectrum: the step-function hazard generates the largest selection within the power law family at fixed $\bar{\Lambda}$ and dramatically reduces non-neutrality relative to Calvo.

The power law family The power law $G(\xi) = (\xi/\bar{\xi})^\gamma$, $\gamma > 0$, spans the spectrum: low γ gives cheap adjustment and a gently rising hazard (near Calvo), high γ concentrates costs near $\bar{\xi}$ and steepens the hazard (near Golosov-Lucas). The induced hazard and cost are:

$$\Lambda(x) = \left(\frac{Bx^2}{w\bar{\xi}} \right)^\gamma, \quad C(\varphi) = \frac{\gamma}{\gamma + 1} w\bar{\xi} \varphi^{1+1/\gamma}. \quad (23)$$

The hazard expression applies for $|x| \leq \sqrt{w\bar{\xi}/B}$; for larger gaps $\Lambda(x) = 1$, defining the inaction band boundary. Λ is continuous but not C^1 at $|x| = \sqrt{w\bar{\xi}/B}$; boundary sets have measure zero under absolutely continuous f_{ss} . This family nests the uniform distribution ($\gamma = 1$) and extends to right-skewed cost distributions ($\gamma < 1$).

Thus γ governs the spectrum: $\gamma = 1$ is uniform costs and quadratic C (Dotsey, King, and Wolman, 1999; Khan and Thomas, 2008); $\gamma \rightarrow \infty$ is the step hazard of Golosov and Lucas (2007); the Calvo approximation arises only in the frequency-calibrated limit of Proposition 9.

The power-law family varies one coordinate of G : the local exponent γ that traces Calvo to Golosov-Lucas. Table 2 summarises the three reference points along it. Placing this axis against a second coordinate, the global tail of G that the local moment leaves free, locates the major price-setting literatures as coordinates on a single primitive.

Table 2 collects the spectrum: G indexes the range from Calvo (dispersed G , vanishing selection wedge) through uniform G to Golosov-Lucas (point mass, step hazard, polar selection), traced continuously by the power-law γ .

Table 2: The state-dependence spectrum

	Calvo	Uniform G	Golosov-Lucas
Distribution G	Dispersed	$U[0, \bar{\xi}]$	Point mass at $\bar{\xi}$
$C(\varphi)$	Highly curved	Quadratic	Linear on $\{0, 1\}$
$C'''(\varphi)$	$\rightarrow \infty$	$w\bar{\xi}$	0 or ∞
Hazard $\Lambda(x)$	$\approx$ constant	$Bx^2/(w\bar{\xi})$	Step function
Selection	$\approx$ none	Moderate	Maximum
NKPC slope	$\approx$ constant	State-dependent	Very steep
Non-neutrality	Maximum	Intermediate	Minimum
Power law γ	$\rightarrow 0^+$	1	$\rightarrow \infty$

Notes: Three reference points on the power law spectrum $G(\xi) = (\xi/\bar{\xi})^\gamma$. The Calvo column is the frequency-calibrated limit $\gamma \rightarrow 0^+$ with $\bar{\xi}(\gamma) \rightarrow \infty$ to hold $\bar{\Lambda}$ fixed (Proposition 9); Uniform G corresponds to $\gamma = 1$; Golosov-Lucas is the polar limit $\gamma \rightarrow \infty$ at fixed $\bar{\xi}$. The hazard, NKPC slope, and non-neutrality rows describe the aggregate consequences of the cost shape: dispersed G produces near-random repricing and near-Calvo aggregates; a point mass at $\bar{\xi}$ produces the Ss -style step hazard of Golosov and Lucas (2007). The “Maximum”/“Minimum” entries in the selection and non-neutrality rows are extremal *within the power-law family* at fixed $\bar{\Lambda}$, not across all hazards (Proposition 8). The empirical location on this spectrum is taken up in Section 4.

Locating the literature in two dimensions Each cost distribution can be summarised as a *generalized power law*, by two coordinates: a local power-law exponent γ at the origin and a tail index θ at infinity. Explicitly, $G(\xi) \sim (\xi/\bar{\xi})^\gamma$ near the origin (so $\nu = 2\gamma$), while the right tail follows the generalized-Pareto form $1 - G(\xi) = (1 + \theta \xi/\sigma)^{-1/\theta}$, whose index runs from bounded support ($\theta = -1$, the uniform case) through exponential decay ($\theta \rightarrow 0$) to heavy polynomial tails ($\theta = 1/\alpha$; $\theta = +\frac{1}{2}$ for Lomax with $\alpha = 2$). The space

is therefore two-dimensional, and [Alvarez, Lippi, and Oskolkov](#)’s one-parameter hazard-exponent spectrum is a one-dimensional slice of it (Figure 1). The horizontal axis is the local exponent γ of G at the origin (equivalently $\nu/2$ in ALO notation): $\gamma \rightarrow 0$ is Calvo, $\gamma = 1$ is the Khan-Thomas / ALO-empirical latitude, $\gamma \rightarrow \infty$ is Golosov-Lucas. The vertical axis is the global tail shape of G : bounded support, exponential decay, heavy polynomial decay. The Calvo–Golosov-Lucas (power-law G) spectrum is a *horizontal line* in this plane: every member of the power-law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ has bounded support $[0, \bar{\xi}]$, so as γ ranges from 0 (Calvo) through 1 (Khan-Thomas) to ∞ (Golosov-Lucas), only the horizontal coordinate moves; the tail-shape coordinate is fixed at *bounded*. [Alvarez, Lippi, and Oskolkov \(2022\)](#)’s local log-log hazard slope ($\nu \approx 2$) pinpoints $\gamma = 1$ (a *vertical line* in the plane) but the local slope alone does not pin down the tail, since it constrains G near the origin rather than its right tail. Their kurtosis–frequency moments do carry tail information, since price-change kurtosis is tail-sensitive; but as an instrument for the tail it is weak, so the points along this vertical line stay close in kurtosis–frequency space while differing materially in the right tail (Section 4.5). The intersection of the Calvo–Golosov-Lucas line and the ALO line is the Khan-Thomas uniform- G point. [Matějka and McKay \(2015\)](#)’s Shannon-RI cost commits to $\gamma = 1$ and an exponential tail, a single *point* on the ALO line. [Gabaix \(2014\)](#)’s sparsity penalty (quadratic cost over the adjustment probability) coincides with the Khan-Thomas point. The Lomax/Pareto specification (Section 4.5) adds a third point on the ALO line at the heavy-polynomial tail. Tradition-by-tradition, the picture is: the menu-cost tradition traces a horizontal line (the power-law family, on the bounded-support latitude); the ALO empirical anchor selects a vertical line (the local exponent); each rational-inattention specification is a single point. The duality framework makes the geometry visible: ALO identifies one coordinate, each information-theoretic axiomatisation commits to a particular tail, and the empirical agenda for disambiguating among the points on the ALO line is the right tail of the price-change distribution. Equivalently, the generalized power law makes [Alvarez, Lippi, and Oskolkov](#)’s hazard-exponent spectrum two-dimensional: the local exponent they identify is the horizontal coordinate, with their spectrum the bounded edge $\theta = -1$, and the tail θ , which the local slope and price-change kurtosis leave free, is the orthogonal coordinate along which transmission nonetheless varies (Section 4.5).

3.4 Rational inattention as a microfoundation of sticky prices

The proposition that rational inattention can microfound sticky prices is established in the literature: [Reis \(2006\)](#), [Maćkowiak and Wiederholt \(2009\)](#), [Stevens \(2020\)](#), and the broader sticky-price rational-inattention tradition (reviewed in [Maćkowiak, Matějka, and Wiederholt](#),

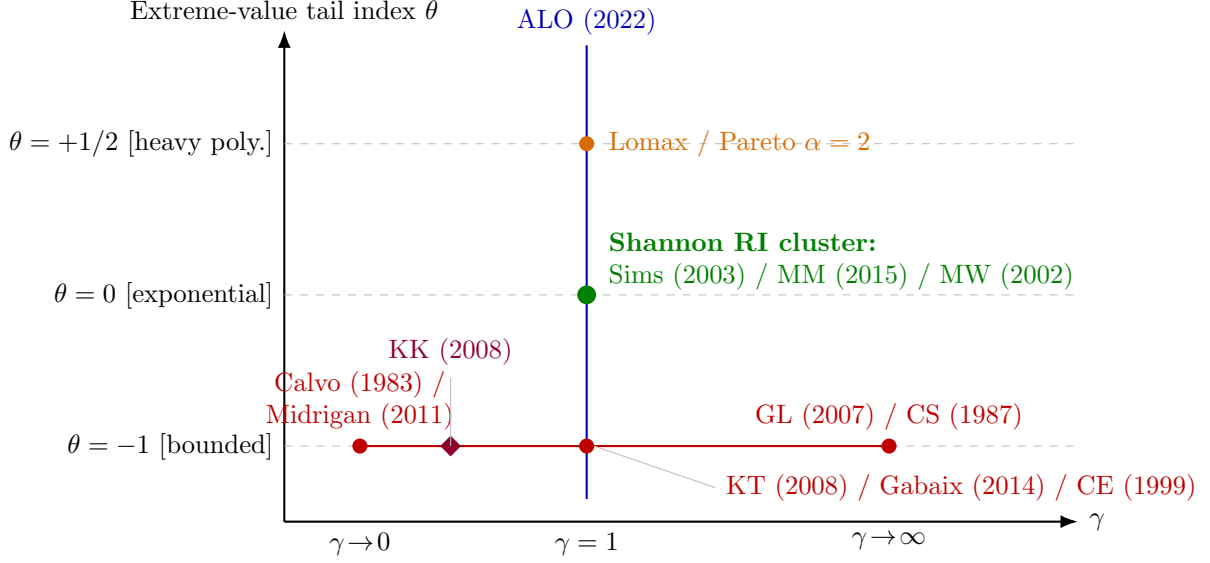


Figure 1: Cost-distribution specifications as a generalized power law (γ, θ) : local hazard exponent γ (horizontal) and tail index θ (vertical). ALO’s hazard-exponent spectrum is the bounded edge $\theta = -1$; the tail is the orthogonal dimension.

Notes: Abbreviations: ALO = Alvarez-Lippi-Oskolkov, KT = Khan-Thomas, GL = Golosov-Lucas, CS = Caplin-Spulber, KK = Klenow-Kryvtsov, CE = Caballero-Engel, MM = Matějka-McKay, MW = Mattsson-Weibull. Horizontal axis: $\gamma = \nu/2$, the local hazard exponent of G at the origin. Vertical axis: θ , the Pickands (1975) tail index ($\theta < 0$ bounded; $\theta = 0$ exponential; $\theta > 0$ polynomial; Lomax $\alpha \Rightarrow \theta = 1/\alpha$). The PL family is the horizontal red line; Alvarez, Lippi, and Oskolkov (2022) the vertical blue line. Their intersection co-locates Khan and Thomas (2008), Gabaix (2014), Caballero and Engel (1999), and (at the GL endpoint) Golosov and Lucas (2007) / Caplin and Spulber (1987). The Shannon-RI cluster at $(1, 0)$ is the binary-RI core: Sims (2003), Matějka and McKay (2015), Mattsson and Weibull (2002). *Off-figure mentions:* Maćkowiak and Wiederholt (2009) (continuous Shannon) and McFadden (1974) (multinomial logit) project to $(\gamma, \theta) = (1, 0)$ only in the binary collapse. Midrigan (2011) (CalvoPlus two-point cost) sits at the Calvo end of the PL line. Klenow and Kryvtsov (2008)’s kurtosis-match pins $\gamma \approx 0.4$; Cavallo (2018)’s scraped tails identify $\theta > 0$ but no specific latitude. Dotsey, King, and Wolman (1999) is an interior PL point with $\gamma < 1$ (no specific value); Cooper and Haltiwanger (2006)’s convex+fixed hybrid has no clean (γ, θ) coordinate. Tsallis (1988), Hofbauer and Sandholm (2002), and Bloedel, Denti, and Pomatto (2025) span γ ; Reis (2006) is time-dependent and off the map.

2023) build models in which the friction is informational rather than physical, with Shannon-style information costs in place of menu costs. The closest antecedent is Woodford (2009), who derives a generalised state-dependent adjustment hazard from an information-processing constraint; the present construction instead reads a distribution of *physical* adjustment costs as attention costs: an exact algebraic duality between the two readings, rather than a derivation of the hazard from a primitive information constraint. More generally, Fosgerau et al. (2020) establish an equivalence between additive random-utility discrete choice and rational inattention with generalised-entropy costs; the binary menu-cost/rational-inattention equiva-

lence here is the cost-draw instance of that class, specialised to a distribution G of adjustment costs and embedded in a dynamic pricing model. For any pricing behaviour rationalised by a menu-cost economy with cost distribution G , there exists an inattention economy with the same G — now read as a distribution of attention costs — that produces the same firm-level decisions, the same cross-section f_{ss} , the same average reset frequency $\bar{\Lambda}$, the same Phillips curve slope $\kappa = (1 + 2\gamma)\bar{\Lambda}$, and the same selection wedge. No modification to the model’s ingredients is required: only the interpretation of the cost distribution changes. The threshold rule “adjust iff $w\xi \leq S$ ” reads as “update the firm’s information set about the current desired price iff the value of doing so exceeds the cognitive cost $w\xi$ ”; the convex cost $C(\varphi)$ on the probability-choice side reads as the shadow price of cognitive resources allocated to price-tracking. The relabelling is exact at the ex-ante extensive margin under the additivity assumption of Section 2; richer information-choice problems in the Mackowiak–Wiederholt vein involve signal-allocation margins beyond the binary update decision that the present equivalence does not span.

RI restatement: the firm’s Bellman equation line by line A firm with current price gap x and an independent cost draw $\xi \sim G$ each period solves

$$V(x) = \mathbb{E}_\xi \left[\max \left\{ S(x) - w\xi + \beta V(x^*), \beta \mathbb{E}[V(x') | x] \right\} \right], \quad (24)$$

where $S(x) = Bx^2$ is the surplus from closing the gap, x^* is the reset target, and $x' | x$ is the evolved gap absent adjustment. The threshold rule “adjust iff $w\xi \leq S(x) + \beta[V(x^*) - \mathbb{E}[V(x') | x]]$ ” follows from the max.

Reading (24) as a rational-inattention model is a term-by-term relabelling: every mathematical object stays put while the verbal interpretation changes. The state x becomes the gap between the firm’s posted price and the desired price implied by its current information set — the same observable, with the cognitive reading that the firm has not yet processed the new p^* . The draw $\xi \sim G$ becomes the realised attention cost of incorporating the new desired price; $w\xi$ is now a cognitive cost, with G a distribution of attention costs. The surplus $S(x) = Bx^2$ becomes the informational value of updating: the payoff lost from operating under stale information. The continuation values $V(x^*)$ and $\mathbb{E}[V(x') | x]$ are the values of next period under updated or stale information respectively. The threshold “ $w\xi \leq S(x) + \dots$ ” reads as “update the information set iff the cognitive cost does not exceed the informational benefit.”

The Fenchel duality of Theorem 1 rewrites the Bellman exactly in the convex form

$$V(x) = \max_{\varphi \in [0,1]} \{ \varphi [S(x) + \beta V(x^*)] + (1 - \varphi) \beta \mathbb{E}[V(x') \mid x] - C(\varphi) \}, \quad (25)$$

with φ the ex-ante choice of adjustment probability and $C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi)$ the dual cost. The interior first-order condition

$$S(x) + \beta [V(x^*) - \mathbb{E}[V(x') \mid x]] = C'(\varphi^*)$$

replaces the threshold rule with a smooth equation in φ^* . Under the RI reading, φ is the probability of *updating* the firm’s information set, $C(\varphi)$ is the shadow price of attention as a function of that probability, and the FOC equates marginal informational benefit with marginal shadow price of attention.

Aggregating across the cross-section of firms, the hazard $\Lambda(x) = G(Bx^2/w)$ (the probability the firm adjusts / updates given the current gap), the stationary distribution $f_{ss}(x)$, the average frequency $\bar{\Lambda} = \int \Lambda(x) f_{ss}(x) dx$, the Phillips slope $\kappa = (1 + 2\gamma)\bar{\Lambda}$ (with selection share $2\gamma/(1 + 2\gamma)$), and the kurtosis-frequency moments of Alvarez, Le Bihan, and Lippi (2016) all carry through unchanged. The interpretation of each object — physical resource cost or cognitive attention cost — is a choice separate from anything micro pricing data alone can adjudicate, but does not affect the model’s predictions for any observable price-setting moment.

Why Shannon mutual information? The rational-inattention literature adopts the Shannon mutual-information cost (in binary form, the Kullback-Leibler divergence over the adjustment probability (Sims, 2003; Maćkowiak and Wiederholt, 2009; Matějka and McKay, 2015)) on three a priori grounds, none empirical: *axiomatic uniqueness* (the only functional satisfying the Shannon-Khinchin axioms), *operational grounding* (channel capacity in Shannon’s communication theory, imported by Sims (2003) as the processing constraint), and *tractability* (logit choice, additivity across decisions, a shadow price λ in nats). The duality gives the choice a structural reading: Shannon cost is the dual of an *exponential* G — one point of the entropy correspondence of Proposition 4. Choosing an attention-cost functional form is therefore equivalent to choosing a menu-cost distribution shape, so “Shannon” is an implicit, and now visible and testable, distributional assumption rather than a free modelling default.

The implication is that the observed price stickiness that the menu-cost literature attributes to physical resource frictions can equally be rationalised by attention frictions. The

empirically identifiable object in micro price data — the cross-section of price changes, the frequency of adjustment, the kurtosis–frequency moments of [Alvarez, Le Bihan, and Lippi \(2016\)](#), the impulse-response patterns of [Midrigan \(2011\)](#) and [Nakamura and Steinsson \(2010\)](#) — is the distribution G (equivalently, the convex cost C). The *interpretation* of G , physical menu cost or cognitive attention cost, is a choice separate from anything micro pricing data alone can adjudicate. From the standpoint of observed price-setting behaviour, the menu-cost and rational-inattention traditions are not competing theories of price stickiness but two readings of a single primitive.

The Calvo–Goloso-Lucas spectrum carries through this re-reading without modification, since the same γ that indexes cost concentration now indexes how sharply firms allocate attention to the size of the gap. Calvo (low γ , dispersed costs, weak selection) is the regime in which firms allocate attention weakly to the size of the gap, updating at roughly random intervals, close to the time-dependent updating of [Reis \(2006\)](#)’s inattentive-producer model. Goloso-Lucas (high γ , concentrated costs, strong selection) is the regime in which firms allocate attention sharply by the size of the gap, updating only when the cognitive value of doing so is large — close to the threshold-attention rules of the [Sims \(2003\)](#) tradition. Intermediate γ traces a continuous family of attention rules between these, so [Maćkowiak and Wiederholt \(2009\)](#) and the broader sticky-price rational-inattention literature correspond to points on the same spectrum, with γ a structural shape parameter of the attention-cost function.

What the two readings predict identically is the period-by-period decision rule; they can differ in how that rule responds to the environment and in welfare. Trend inflation is the natural test: under the menu-cost reading G is regime-independent, whereas under inattention G can shift as firms re-allocate attention towards inflation-relevant signals, so part of the dispersion–inflation co-movement of [Vavra \(2014\)](#) could reflect endogenous attention rather than the gap-distribution channel alone. And $C(\varphi^*)$ is an expected physical resource cost under one reading and an attention cost under the other: not commensurable, and not separated by price-change moments. Distinguishing the two is a structural choice, to be settled by direct evidence on the margin firms find costly, not by the pricing moments alone.

The duality thus recasts a two-decade question — whether observed price stickiness reflects physical menu costs or rational inattention — as one without an observational answer at the level of price-setting behaviour: both traditions identify the same G , and what separates them is only what they take G to measure.

3.5 Aggregate inflation and the Phillips curve

Because $\Lambda(x) = G(Bx^2/w)$ is closed-form, aggregate inflation is a direct functional of G . A repricing firm closes its gap, changing its log price by $-x$, so with $f_t(x)$ the gap density,

$$\pi_t = -E_{f_t}[\Lambda(x_t) \cdot x_t] = - \int \Lambda(x) x f_t(x) dx. \quad (26)$$

To understand how aggregate inflation responds to nominal disturbances, consider a small aggregate shock μ_t that shifts the desired price of all firms by μ_t , moving the cross-sectional distribution of gaps from the steady-state $f_{ss}(x)$ to $f_{ss}(x + \mu_t)$.

Proposition 10 (Nonlinear Phillips curve).

Under the quadratic surplus approximation $S(x) = Bx^2$ and menu cost distribution G , aggregate inflation satisfies

$$\pi_t = -E_t \left[G \left(\frac{Bx_t^2}{w} \right) \cdot x_t \right]. \quad (27)$$

For small aggregate disturbances around the steady state with gap distribution f_{ss} :

$$\pi_t \approx \kappa(G) \cdot \mu_t, \quad (28)$$

where the distribution-dependent slope is

$$\kappa(G) := \underbrace{\int G \left(\frac{Bx^2}{w} \right) f_{ss}(x) dx}_{\text{extensive margin (frequency)}} + \underbrace{\int g \left(\frac{Bx^2}{w} \right) \frac{2Bx^2}{w} f_{ss}(x) dx}_{\text{intensive margin (selection)}}. \quad (29)$$

Proof.

See Online Appendix C. ■

The extensive margin (first term) is the average repricing frequency, the only margin under Calvo; the intensive margin (second term) is selection, absent under Calvo and dominant under Golosov-Lucas. The slope is a partial-equilibrium one-time translation of f_{ss} ; general-equilibrium feedbacks modify it quantitatively, complementing [Alvarez, Le Bihan, and Lippi \(2016\)](#).

Corollary 4 (How G governs the Phillips curve).

Specialising the slope $\kappa(G)$ in (29) to the canonical menu-cost distributions shows how the balance between the two margins shifts as G concentrates: dispersed G leaves only the frequency term, while concentrated G makes selection dominate. The three benchmark cases are:

- (i) **Calvo limit:** When G is highly dispersed, g is small across the region where f_{ss} has mass, and the intensive margin is negligible. The slope approaches $\kappa \approx \bar{\Lambda}$, approximating the textbook New Keynesian Phillips Curve.
- (ii) **Golosov-Lucas:** When G is degenerate, $G(Bx^2/w) = \mathbf{1}[|x| \geq \bar{x}]$ and $g = \delta_{\bar{x}}$. The intensive margin dominates: inflation is driven by selection of firms at the threshold. The slope is very steep, implying low non-neutrality.
- (iii) **Uniform:** When $G = U[0, \bar{\xi}]$, the slope simplifies to

$$\kappa^U = \frac{3B}{w\bar{\xi}} E_{ss}[x^2], \quad (30)$$

the second moment of the gap distribution, which reduces to $\frac{3B}{w\bar{\xi}} \text{Var}_{ss}(x)$ when gaps are mean-zero (the zero-trend-inflation benchmark) and otherwise also reflects the mean drift. The slope depends on the steady-state dispersion of price gaps. Economies with more dispersed gaps have steeper Phillips curves, because the selection margin is stronger when more firms have large gaps near the threshold.

- (iv) **Power law γ :** The slope κ is strictly increasing in γ . Higher γ means more concentrated menu costs, a steeper hazard, stronger selection, and a steeper Phillips curve.

3.6 Non-neutrality decomposition

Non-neutrality is inversely related to the slope κ , which decomposes as $\kappa(G) = \bar{\Lambda} + (\kappa - \bar{\Lambda})$, frequency plus selection wedge: how often firms reprice, plus how well repricing is targeted. The wedge is zero under Calvo and strictly positive for any G with mass where Λ is increasing. For the power-law family the decomposition is exact,

$$\kappa = (1 + 2\gamma)\bar{\Lambda}, \quad \text{selection wedge} = 2\gamma\bar{\Lambda}, \quad (31)$$

so the ratio $\kappa/\bar{\Lambda} = 1 + 2\gamma$ and the selection share $2\gamma/(1 + 2\gamma)$ depend only on γ , not on f_{ss} , B , or w . Three consequences follow (formal statements and proofs in Online Appendix C): non-neutrality is strictly decreasing in γ , largest at Calvo ($\gamma \rightarrow 0$) and smallest at Golosov-Lucas ($\gamma \rightarrow \infty$); two economies with the *same* repricing frequency can differ in non-neutrality, the difference being entirely the selection wedge; so frequency alone does not determine non-neutrality — the shape of G does. The dynamic counterpart (Online Appendix D) shows low γ giving near-random repricing and large persistent output effects, high γ targeted repricing and small transient effects.

4 Quantitative analysis

The theoretical results of Sections 2–3 make G the structural primitive governing adjustment dynamics. This section records a compact calibration of the cost-shape parameter γ and the single illustration of the shared cross-walk calibration that the introduction promised (Section 4.3). It then solves the full dynamic model globally across the spectrum (Section 4.5). The smooth first-order condition the duality supplies lets one non-linear solver run continuously from the Calvo to the Golosov-Lucas endpoint.

4.1 A testable prediction: hazard-based identification

The hazard’s structural anatomy $\nu = 2\gamma$ (Section 3.2) doubles as an empirical-identification route. For the power-law family, taking logs of $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$ gives

$$\log \Lambda(x) = 2\gamma \log |x| + \text{const.} \quad (32)$$

so the slope of $\log \Lambda$ against $\log |x|$ recovers γ from the hazard shape alone, independently of $\bar{\xi}$, B , and w . A Monte Carlo confirms recovery to mean absolute error 0.012 and $R^2 > 0.99$ (Online Appendix D).

Implementing this on actual data requires observing the price gap x at each decision point, which is not recorded in standard micro price datasets (scanner or CPI) — only price changes Δp . Model-based proxies for x exist but are not pursued here. Datasets with posted reference prices or shelf tags permit constructing x directly; that is the natural setting for the hazard-based test. The remainder of this section uses moment matching to obtain an illustrative calibrated value of γ , not a fully identified estimate.

4.2 Calibration and moment matching

The baseline calibration sets $B = w = 1$ (normalisation) and $\sigma_z = 0.04$ (quarterly standard deviation of idiosyncratic desired-price shocks, consistent with Golosov and Lucas 2007 and Nakamura and Steinsson 2008). For each γ , the upper bound $\bar{\xi}$ is calibrated to match a quarterly repricing frequency of 24%, corresponding to approximately 8–9% monthly excluding temporary sales (Nakamura and Steinsson, 2008); under constant monthly hazards, $1 - (1 - 0.085)^3 \approx 0.234$. Throughout, φ is interpreted as a per-period probability.

In the baseline model with homogeneous σ_z , simulated kurtosis ranges from 1.33 (at $\gamma = 2$) to 2.79 (at $\gamma = 0.15$) across the γ grid (Online Appendix D) — well below the empirical kurtosis of 4–6, a well-known limitation of single-product menu cost models with

a homogeneous shock process. Following [Midrigan \(2011\)](#), the extended model draws per-product σ_z from a lognormal distribution with mean 0.04 and coefficient of variation $CV = 1$. The CV is imposed, not estimated; this generates kurtosis in the empirical range while preserving the identification of γ from the hazard shape. Sensitivity to $CV \in \{0.5, 1.5\}$ is left for future quantitative work.

4.3 One illustration: a shared calibration

The cross-walk lets a single estimate calibrate both readings of the cost distribution at once. The power-law hazard $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$ has $\log \Lambda$ linear in $\log |x|$ with slope 2γ , so [Alvarez, Lippi, and Oskolkov \(2022\)](#)’s heterogeneity-corrected hazard exponent $\nu \approx 2$ reads as $\hat{\gamma} = \nu/2 \approx 1$ — the Khan-Thomas uniform- G point, equivalently a quadratic attention cost on the rational-inattention side. A purely moment-matching alternative that targets the raw price-change kurtosis ≈ 5 in the heterogeneous- σ_z model gives $\hat{\gamma} \approx 0.4$. At $\hat{\gamma} = 1$ the aggregate slope $\kappa = (1+2\gamma)\bar{\Lambda}$ assigns selection $2\gamma/(1+2\gamma) = \frac{2}{3}$ of the slope; at $\hat{\gamma} = 0.4$, 44%. The selection share is an analytic function of γ alone, independent of the targeted frequency, spanning 23% to 75% over the literature-relevant range $\gamma \in [0.15, 1.5]$. This is an illustration of the cross-walk, not its point: the value is that one number from price-change microdata disciplines two cost specifications the literature has treated as unrelated, whatever share it implies.

4.4 Hazard functions and impulse responses

The simulated repricing hazard across the spectrum (Online Appendix D) is concave in $|x|$ for $\gamma < 1$ and convex for $\gamma > 1$, with the benchmark $\hat{\gamma} = 1$ at the knife-edge producing a quadratic hazard $\Lambda(x) \propto x^2$. The output response to a 1% monetary shock (Online Appendix D), holding the repricing frequency fixed across γ , shows state-dependent economies converging faster than Calvo as repricing targets the most-mispriced firms: the cumulative output gap is $\approx 37\%$ of the Calvo path at $\hat{\gamma} = 1$ and $\approx 75\%$ at $\hat{\gamma} = 0.4$. A robustness exercise on Dominick’s scanner data ([Midrigan, 2011](#); [Chevalier, Kashyap, and Rossi, 2003](#)) (Online Appendix E) yields higher frequency and kurtosis than CPI, consistent with sectoral heterogeneity in G paralleling [Oskolkov and Lippi \(2026\)](#)’s plant-level investment evidence; modelling sector-specific $(\gamma_s, \bar{\xi}_s)$ is a natural next step.

4.5 Global non-linear solution of the dynamic spectrum

The smooth first-order condition $S = C'(\varphi)$ and the C^1 value function (Theorem 2) make the full dynamic model of Section 3.1 solvable by a single global non-linear method — the repeated transition method (RTM) of Lee (2026). This method runs continuously across the entire spectrum, including the Golosov-Lucas endpoint, where the ex-ante value acquires a corner that linearised and perturbation methods cannot accommodate. The exercise serves two ends: a credibility check that the closed-form decomposition of Section 3 survives as a tight numerical approximation in the unapproximated dynamic model; and a comparison across the tail of G (below) showing that the right tail, which the local hazard exponent does not identify, shapes dynamic transmission.

It is worth pausing on the model used here versus the model used in the calibration, since the two differ deliberately in how the cross-section is kept non-degenerate. The CPI moment exercise of Section 4.2 includes idiosyncratic price shocks ($\sigma_z = 0.04$ with cross-product CV = 1), which generate the kurtosis used by the kurtosis-matching alternative. The global solution below instead sets $\sigma_z = 0$ and lets trend inflation alone keep the cross-section non-degenerate. This choice is made both because the differentiability and decomposition results of this paper do not require idiosyncratic smoothing, and because the cleaner setting isolates the analytical content the global exercise is meant to check. The parameter γ is the same object in both: it is the shape of the cost distribution G , and the hazard $\Lambda(x) = G(Bx^2/w) = (Bx^2/(w\bar{\xi}))^\gamma$ is identified from the gap-conditional probability of adjustment regardless of which mechanism keeps f_{ss} non-degenerate. The benchmark $\hat{\gamma} = 1$ from the hazard-exponent route is the value plugged into the global solution.

The smooth FOC-envelope structure of Section 3.1 is implemented directly, with each period tracing both the value V and its gradient V_x so that the surplus and the resetter's optimum can be read off without enumeration. The matched conditional expectations of V' and V'_x deliver the surplus $S = V^a - V^n$, hence the hazard $\Lambda = G(S/w)$, and the resetter's first-order condition, with V_x updated by the envelope $V_x = (1 - \Lambda)V_x^n$. No individual-level grid search for x'^* is required, since the resetter's optimum is pinned down by the first-order condition rather than by enumerating candidate reset gaps. The scalar summary that drives the matching is the lagged price-dispersion index Δ_{t-1} ; the full routine and a within-exogenous-state monotonicity check of $V(x; X)$ in Δ_{t-1} are in Online Appendix G.

Relative to the literature: Golosov and Lucas (2007), Midrigan (2011), and Nakamura and Steinsson (2010) solve calibrated menu-cost models at specific points of the spectrum, typically with linearised aggregate dynamics, and Alvarez, Le Bihan, and Lippi (2016) summarise non-neutrality through a first-order-perturbation sufficient statistic. Among related nonlinear solutions, Costain and Nakov (2011) and Costain, Nakov, and Petit (2022) solve

generalised-hazard menu-cost models with a fully nonlinear firm-level distribution but linearised aggregate dynamics (the Reiter method), and [Alexandrov \(2020\)](#) and [Blanco et al. \(2024\)](#) solve nonlinear state-dependent models at specific calibrations; the global solution here is nonlinear in the aggregate dynamics as well.

Consistency with the analytical decomposition The static decomposition is exact under two conditions stated in Section 3: (i) the repricing surplus is quadratic in x , $S(x) = Bx^2$, so the hazard takes the power-law form $\Lambda = (Bx^2/(w\bar{\xi}))^\gamma$; (ii) the stationary gap distribution f_{ss} is supported in the interior of the inaction band, so $\kappa = (1 + 2\gamma)\bar{\Lambda}$ holds without a boundary correction. In the dynamic model neither condition holds exactly (the dynamic surplus is only approximately quadratic, and the support of f_{ss} is the inaction band itself rather than its strict interior), so the analytical decomposition becomes a numerical approximation that the global solution lets me evaluate. The global exercise also makes a third, analytical, property useful: (iii) the ratio $\kappa/\bar{\Lambda} = 1 + 2\gamma$ does not depend on the curvature B of S (Online Appendix C), so any discrepancy between the static and dynamic B leaves $\kappa/\bar{\Lambda}$ untouched. I check (i)–(ii) numerically and exploit (iii) across the spectrum γ , with $\bar{\xi}(\gamma)$ recalibrated to hold the average repricing frequency at $\bar{\Lambda} = 0.10$.

(i) The dynamic surplus is approximately quadratic near the optimum, so the hazard retains its power-law form to that approximation. Its local curvature is, however, roughly four times the static $\varepsilon/2$, reflecting the discounted accumulation of flow losses over the firm’s expected mispricing duration: the static B misses the level of curvature, but not its shape. The closed-form $\kappa = (1 + 2\gamma)\bar{\Lambda}$ in the dynamic model is therefore valid as a tight approximation, not as an exact identity.

(ii) Mass of f_{ss} at the $\Lambda = 1$ boundary is below 10^{-5} throughout the spectrum, so the boundary-correction term in κ is numerically negligible.

(iii) Both κ and $\bar{\Lambda}$ scale with B through the hazard, so the larger dynamic curvature drops out of the ratio. The selection share $2\gamma/(1 + 2\gamma)$ rises from 17% at $\gamma = 0.1$ to 86% at $\gamma = 3$. At the benchmark $\hat{\gamma} = 1$ it equals $\frac{2}{3} \approx 67\%$, the headline value of the calibration above, intact to the approximation error in (i).

The cross-sectional welfare cost of adjustment $\int C(\Lambda(x)) f_{ss}(x) dx$ rises with γ and saturates: stronger selection buys lower dispersion at a higher resource cost. Figure 2 collects these objects.

Across the local-exponent dimension The global solver traces transmission along both axes of the two-dimensional map. Along the horizontal axis, Figure 3 compares three bounded-support power-law specifications — $\gamma = 0.4$ (dispersed, Calvo-leaning), $\gamma = 0.5$,

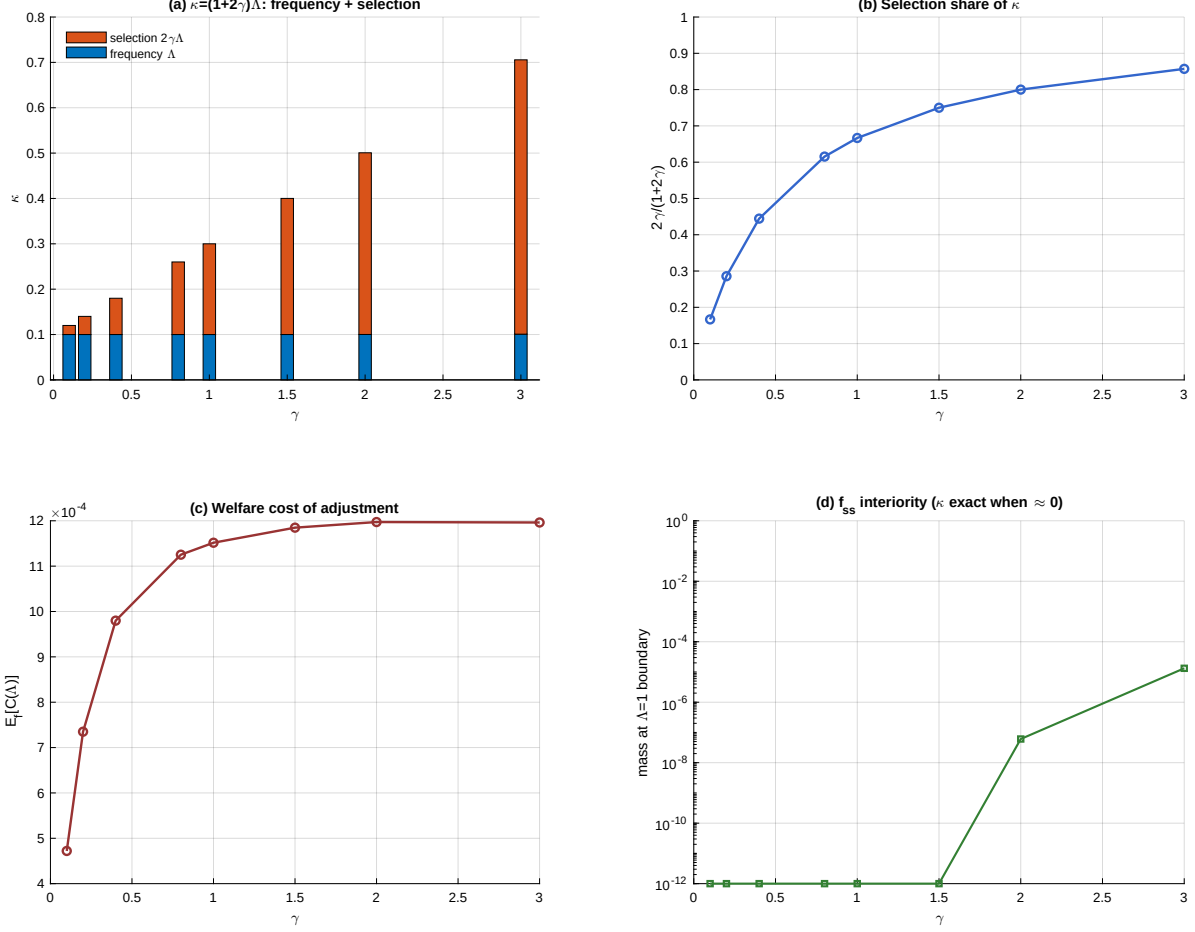


Figure 2: Steady-state spectrum comparative static across γ

Notes: Steady-state objects across the spectrum index γ , with $\bar{\xi}(\gamma)$ recalibrated to hold the average repricing frequency at $\bar{\Lambda} = 0.10$. (a) the Phillips slope $\kappa = (1+2\gamma)\bar{\Lambda}$ decomposed into the frequency component $\bar{\Lambda}$ and the selection wedge $2\gamma\bar{\Lambda}$; (b) the selection share $2\gamma/(1+2\gamma)$ ($\frac{2}{3} \approx 67\%$ at the benchmark $\gamma = 1$); (c) the welfare cost of adjustment $\int C(\Lambda(x)) f_{ss}(x) dx$; (d) the mass of f_{ss} at the inaction-band boundary, negligible throughout, so the boundary correction to $\kappa = (1+2\gamma)\bar{\Lambda}$ is numerically negligible.

and $\gamma = 1$ (Khan-Thomas) — each globally solved and calibrated to $\bar{\Lambda} = 0.10$. As γ rises the selection-margin response strengthens monotonically and the output response shrinks, exactly the steepening of $\kappa = (1+2\gamma)\bar{\Lambda}$ now traced out in the full non-linear general equilibrium. Unlike the tail, this axis is *locally identified*: the hazard slope $\nu = 2\gamma$ reads γ off the data directly, so the variation in transmission it produces is in principle recoverable from the price-change distribution.

Across the tail dimension Holding the local exponent fixed, the same global solver now isolates the role of the *tail* of G . Figure 4 compares three specifications — uniform G ($\gamma = 1$, Khan-Thomas), exponential G (Shannon RI), and Lomax/Pareto G ($\theta = +\frac{1}{2}$,

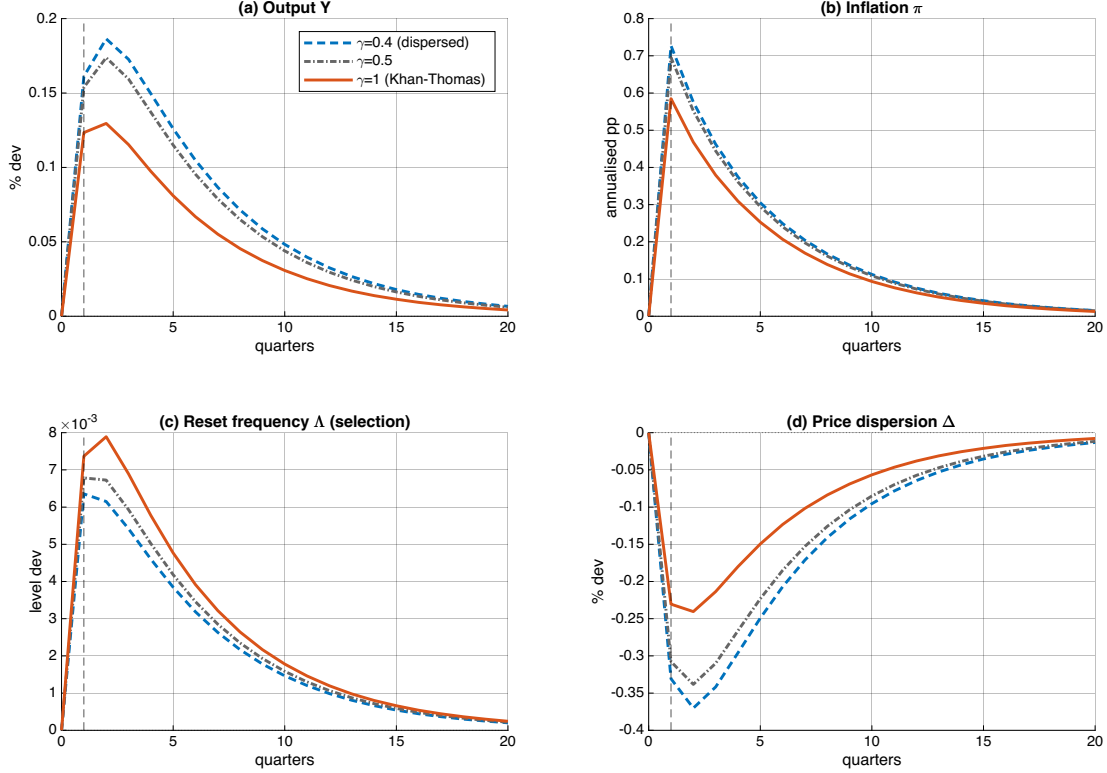


Figure 3: Dynamic transmission across the local exponent γ at fixed (bounded-support) tail ($\bar{\Lambda} = 0.10$)

Notes: Generalised impulse responses to a +1 s.d. demand shock under three bounded-support power-law G specifications ($\theta = -1$ throughout), all calibrated to $\bar{\Lambda} = 0.10$: $\gamma = 0.4$ (dispersed; dashed blue), $\gamma = 0.5$ (dash-dot grey), $\gamma = 1$ (Khan-Thomas; solid orange). Panels (a)–(d): regime-averaged output, inflation, reset frequency $\bar{\Lambda}$ (selection margin), and price dispersion Δ . Selection (panel c) rises monotonically in γ while the output response (panel a) shrinks, tracing the steepening of $\kappa = (1 + 2\gamma)\bar{\Lambda}$. This axis is identified by the local hazard slope $\nu = 2\gamma$, in contrast with the tail dimension of Figure 4.

i.e. $1 - G(\xi) \sim \xi^{-2}$: finite mean but infinite variance) — with the same local exponent $\nu \approx 2$ and the same average frequency $\bar{\Lambda} = 0.10$, differing only in the right tail. The tail acts through the same curvature channel that drives Proposition 3: a heavier right tail thins the density of marginal types at high adjustment probabilities, so the induced cost is more curved there ($C''(\varphi) = w/g(G^{-1}(\varphi))$ rises faster above $\bar{\Lambda}$) and the heavy-tailed Lomax specification produces a markedly stronger response: its cumulative output response (the model analogue of the cumulative impulse response) is about 50% larger than under the uniform or exponential specifications, and its selection-margin impulse about 24% larger, while the bounded (uniform) and exponential specifications — the lighter tails — nearly coincide. This dimension is invisible to the calibrated moments: the local hazard slope ν reads only the origin, and the price-change kurtosis is near-flat across the three (next

paragraph), so a model matched to $(\nu, \bar{\Lambda}, \text{kurtosis})$ does not pin down the selection dynamics, visible in Figure 4, where the impulse responses differ across all four aggregates. The tail is thus a *locally unidentified* dimension, the vertical axis of the two-dimensional map (Figure 1), along which the selection dynamics, and hence transmission, vary even as the local moments do not.

Relation to the kurtosis–frequency sufficient statistic The kurtosis–frequency sufficient statistic of Alvarez, Le Bihan, and Lippi (2016) might seem to recover the tail, since price-change kurtosis is tail-sensitive. Across these three specifications it does not: kurtosis is near-flat — 3.00 (uniform), 3.17 (exponential), 3.00 (Lomax), all close to the Gaussian value — so at the common $\bar{\Lambda} = 0.10$ it does not separate them. The point is sharpest for the uniform and Lomax specifications, which share *both* the local exponent ($\nu \approx 2$) *and* the price-change kurtosis (3.00), hence an identical kurtosis–frequency sufficient statistic, yet generate visibly different impulse responses (Figure 4): same ν , same kurtosis, different transmission. This is consistent with Alvarez, Le Bihan, and Lippi (2016), whose pair is a first-order sufficient statistic in the small-idiosyncratic-shock, low-inflation environment for which it was derived; here the cross-section is held open by trend inflation alone ($\sigma_z = 0$, at a 2% annual target), so its dispersion is drift- rather than diffusion-generated, and the kurtosis a calibration can read is a weak instrument for the tail, just as it weakly identifies γ in the scanner exercise of Online Appendix E. Neither the local hazard slope nor kurtosis disambiguates the right tail; that dimension is read directly from G , not recovered from a price-change moment.

5 Discussion

Further applications Beyond pricing (and investment, in Online Appendix B), the duality applies to any extensive-margin decision with heterogeneous costs. Heterogeneous entry costs micro-found smooth aggregate entry: the free-entry condition $S = wG^{-1}(\varphi)$ equates the marginal entrant’s cost to the marginal cost of raising the entry rate. Heterogeneous vacancy-creation costs micro-found convex posting costs (Pissarides, 2000), with C'' reflecting the cost distribution rather than a technology. Heterogeneous participation costs generate smooth labour supply without Rogerson (1988) lotteries, with elasticity $g(G^{-1}(\varphi))/w$. The same logic extends to technology adoption and credit default (Bernanke, Gertler, and Gilchrist, 1999).

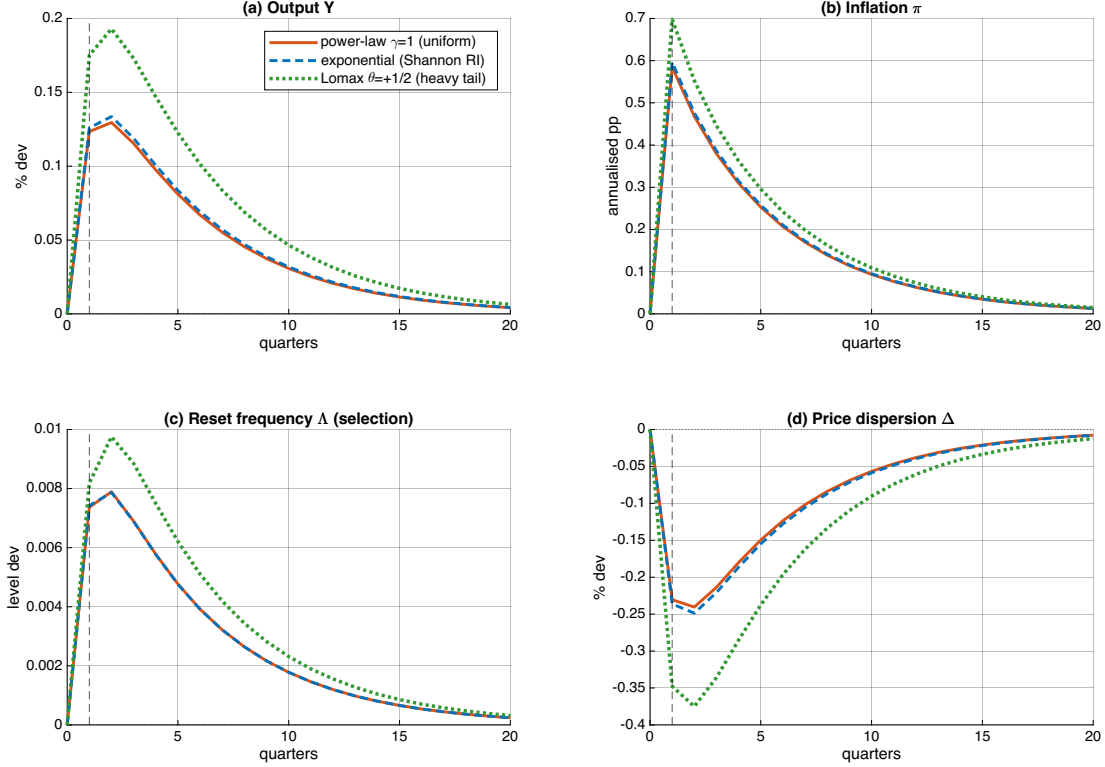


Figure 4: Dynamic transmission across the tail of G at fixed local exponent ($\bar{\Lambda} = 0.10$)

Notes: Generalised impulse responses to a +1 s.d. demand shock under three G specifications in the locally- $\nu = 2$ class, all calibrated to $\bar{\Lambda} = 0.10$ (kurtosis is a model outcome, not a target): power-law $\gamma = 1$ (uniform / Khan-Thomas; solid orange), exponential G (Shannon-RI dual; dashed blue), Lomax/Pareto $\theta = +\frac{1}{2}$ (heavy tail; dotted green). Panels (a)–(d): regime-averaged output, inflation, reset frequency $\bar{\Lambda}$ (selection margin), and price dispersion Δ . The responses differ across all four aggregates; the heavy-tailed Lomax specification departs most visibly on the selection margin (panel c, +24%) and amplifies output (cumulative output $\sim 50\%$ larger than the uniform and exponential specifications that share its ν and kurtosis), while the bounded (uniform) and exponential specifications nearly coincide. Price-change kurtosis is near-flat across the three (uniform 3.00, exponential 3.17, Lomax 3.00): uniform and Lomax share the *same* kurtosis yet differ in transmission, so at common $\bar{\Lambda}$ neither the local-hazard slope nor kurtosis pins down the tail or the selection dynamics within the class.

Friction-extended duality and matching applications When an aggregate friction F enters the surplus per attempted action, the dual rescales as $\tilde{G}(z) = G(Fz)$, $\tilde{C}(\varphi) = C(\varphi)/F$, with FOC $F S = C'(\varphi)$; search-and-matching is the canonical case, $F = F(\theta)$ the matching probability (Pissarides, 2000). When F responds to the aggregate adjustment rate, a Pigouvian wedge $\tau^*(\varphi) = -(dF/d\varphi) S \varphi$ decentralises the planner, collapsing to the Hosios (1990) condition under Cobb–Douglas matching (and its Mangin and Julien (2021) extension otherwise). The efficiency results are known; the contribution is representational — the dual reaches general-equilibrium matching frictions without re-derivation (Online Appendix H).

Structural discipline and identifiability The G framing makes distributional assumptions transparent: quadratic costs require uniform G ; iso-elastic costs require power law G . It supplies a common language: the Calvo–Golosov–Lucas debate, selection in non-neutrality, and the [Khan and Thomas \(2008\)](#) irrelevance result are all statements about G . It also supplies an identifiable primitive, since G is fully recovered from the adjustment hazard (Section 4.1) and disciplined by price-change size distributions and investment spike frequencies (model-dependent). [Oskolkov and Lippi \(2026\)](#) recover G from plant-level data, where the duality implies the same G simultaneously determines C , the investment hazard, and capital misallocation. The aggregate smoothness emphasised by [Auclert, Rognlie, and Straub \(2020\)](#) has an individual-level structural counterpart here: whenever G has a positive density, each agent’s problem is smooth even though realised actions remain binary. The cross-sectional aggregation step is therefore separate from the source of the individual-level smoothness.

Scope and limitations The equivalence is an identity of value functions and adjustment probabilities, not of welfare, interpretation, or causality. The identifiable object is G (equivalently C), not the “type” of friction. Random-menu-cost and probability-choice economies make different predictions about how G responds to changes in the environment; endogenising G is a natural next step. The global solution of Section 4.5 keeps the cross-section non-degenerate by trend inflation alone. Adding idiosyncratic price shocks would change the calibrated curvature B but, by the curvature-invariance of $\kappa/\bar{\Lambda} = 1 + 2\gamma$, would not change the frequency–selection decomposition itself. The methodological claims of Section 4.5 are stated relative to the existing literature surveyed there.

A scope condition worth flagging concerns the additive-separability assumption of Section 2. The duality is exact when the cost realisation ξ enters only the per-period payoff and does not feed back into the state, as in the pricing application (menu costs do not affect the firm’s future gap dynamics) and the Khan–Thomas investment application (ξ does not enter k'). In settings where ξ depletes a stock that itself enters the continuation — most naturally on the household budget side, where paying ξ reduces financial wealth and thereby alters the next-period marginal utility of consumption — the duality identifies expected payoffs and choice probabilities but not state dynamics. The two formulations then diverge in how the cost is paid: the FC formulation has the agent paying $w\xi$ stochastically (only the realised draw, only when adjusting), while the PC formulation has the agent paying $C(\varphi^*)$ deterministically. By construction the two expected outlays match, but the period-by-period wealth processes differ, which matters in representative-agent settings without insurance over the cost draw. Characterising the conditions under which a stock-feedback analogue of the

duality could be recovered is a natural next step.

6 Conclusion

The equivalence established in this paper changes how one should think about adjustment frictions. The distinction between fixed and convex costs is not a distinction between two kinds of economic friction. It is a distinction between two representations of the same friction, linked ex-ante by Fenchel conjugacy. The fixed cost distribution G is the structural primitive that governs both.

Three implications follow, one for theory, one for measurement, and one for the pricing application. First, the lumpy–smooth divide dissolves at the ex-ante extensive margin. Realised actions remain binary at the micro level; what becomes smooth and convex is the ex-ante extensive margin, whenever fixed costs are absolutely continuous with positive density. The deterministic fixed cost, the textbook case, is the singular exception, the one configuration in which smoothness fails. This recasts the standard view that convex costs approximate the “true” lumpy friction: at the ex-ante extensive margin, the deterministic fixed cost is the knife-edge limit and arbitrarily close perturbations restore strict convexity. The same observation provides an independent partial-equilibrium smoothness channel, distinct from the general-equilibrium aggregation debate of [Khan and Thomas \(2008\)](#): with uniform fixed costs, each firm’s ex-ante extensive margin is already convex, so aggregation is not the source of smoothness.

Second, the same mapping $G \mapsto C$ shows that the cost specification in each of menu costs, lumpy investment, discrete choice, rational inattention, and control costs is an implicit distributional assumption on G — one that can be made explicit, tested, and recovered from cost-distribution data. Under the fixed-cost primitive, $C(\varphi^*)$ acquires a concrete resource-cost interpretation, and the smooth first-order condition $S = C'(\varphi)$ replaces the threshold-tracking machinery that standard menu-cost implementations require.

Third, in the pricing application the equivalence gives the repricing hazard $\Lambda(x) = G(Bx^2/w)$ in closed form, from which aggregate non-neutrality decomposes analytically into frequency and selection.

This paper opens a research agenda the duality and global solution put in reach: endogenising G as a menu-cost technology that responds to the inflation environment, with connections to [Vavra \(2014\)](#)’s evidence on the co-movement of price-change dispersion and inflation; and sector-level identification of the cost-shape parameter from sector-specific micro moments.

References

- Alexandrov, Andrey. 2020. “The Effects of Trend Inflation on Aggregate Dynamics and Monetary Stabilization.” CRC TR 224 Discussion Paper 216, University of Bonn and University of Mannheim.
- Alvarez, Fernando, Hervé Le Bihan, and Francesco Lippi. 2016. “The Real Effects of Monetary Shocks in Sticky Price Models: A Sufficient Statistic Approach.” *American Economic Review* 106 (10):2817–2851.
- Alvarez, Fernando, Francesco Lippi, and Aleksei Oskolkov. 2022. “The Macroeconomics of Sticky Prices with Generalized Hazard Functions.” *The Quarterly Journal of Economics* 137 (2):989–1038.
- Auclert, Adrien, Matthew Rognlie, and Ludwig Straub. 2020. “Micro Jumps, Macro Humps: Monetary Policy and Business Cycles in an Estimated HANK Model.” Working Paper 26647, National Bureau of Economic Research.
- Baley, Isaac and Andrés Blanco. 2021. “Aggregate Dynamics in Lumpy Economies.” *Econometrica* 89 (3):1235–1264.
- Benveniste, L. M. and J. A. Scheinkman. 1979. “On the Differentiability of the Value Function in Dynamic Models of Economics.” *Econometrica* 47 (3):727–732.
- Bernanke, Ben S., Mark Gertler, and Simon Gilchrist. 1999. “The Financial Accelerator in a Quantitative Business Cycle Framework.” In *Handbook of Macroeconomics*, vol. 1. Elsevier, 1341–1393.
- Blanco, Andrés, Corina Boar, Callum J. Jones, and Virgiliu Midrigan. 2024. “Non-Linear Inflation Dynamics in Menu Cost Economies.” Working Paper 32094, National Bureau of Economic Research.
- Bloedel, Alexander W., Tommaso Denti, and Luciano Pomatto. 2025. “Modeling Information Acquisition via f -Divergence and Duality.” Working paper, UCLA, NYU Stern, and Caltech. Working paper, October 2025.
- Caballero, Ricardo J. and Eduardo M. R. A. Engel. 1993. “Microeconomic Rigidities and Aggregate Price Dynamics.” *European Economic Review* 37 (4):697–711.
- . 1999. “Explaining Investment Dynamics in U.S. Manufacturing: A Generalized (S, s) Approach.” *Econometrica* 67 (4):783–826.

- . 2007. “Price Stickiness in Ss Models: New Interpretations of Old Results.” *Journal of Monetary Economics* 54 (Supplement 1):100–121.
- Calvo, Guillermo A. 1983. “Staggered Prices in a Utility-Maximizing Framework.” *Journal of Monetary Economics* 12 (3):383–398.
- Caplin, Andrew S. and Daniel F. Spulber. 1987. “Menu Costs and the Neutrality of Money.” *Quarterly Journal of Economics* 102 (4):703–725.
- Carroll, Christopher D. 2006. “The Method of Endogenous Gridpoints for Solving Dynamic Stochastic Optimization Problems.” *Economics Letters* 91 (3):312–320.
- Cavallo, Alberto. 2018. “Scraped Data and Sticky Prices.” *Review of Economics and Statistics* 100 (1):105–119.
- Chevalier, Judith A., Anil K. Kashyap, and Peter E. Rossi. 2003. “Why Don’t Prices Rise During Periods of Peak Demand? Evidence from Scanner Data.” *American Economic Review* 93 (1):15–37.
- Cooper, Russell W. and John C. Haltiwanger. 2006. “On the Nature of Capital Adjustment Costs.” *Review of Economic Studies* 73 (3):611–633.
- Costain, James, Anton Nakov, and Borja Petit. 2022. “Flattening of the Phillips Curve with State-Dependent Prices and Wages.” *The Economic Journal* 132 (642):546–581.
- Costain, James S. and Anton Nakov. 2011. “Distributional Dynamics under Smoothly State-Dependent Pricing.” *Journal of Monetary Economics* 58 (6):646–665.
- Dotsey, Michael, Robert G. King, and Alexander L. Wolman. 1999. “State-Dependent Pricing and the General Equilibrium Dynamics of Money and Output.” *Quarterly Journal of Economics* 114 (2):655–690.
- Fosgerau, Mogens, Emerson Melo, André de Palma, and Matthew Shum. 2020. “Discrete Choice and Rational Inattention: A General Equivalence Result.” *International Economic Review* 61 (4):1569–1589.
- Gabaix, Xavier. 2014. “A Sparsity-Based Model of Bounded Rationality.” *Quarterly Journal of Economics* 129 (4):1661–1710.
- Golosov, Mikhail and Robert E. Lucas, Jr. 2007. “Menu Costs and Phillips Curves.” *Journal of Political Economy* 115 (2):171–199.

- Gourio, François and Anil K. Kashyap. 2007. “Investment Spikes: New Facts and a General Equilibrium Exploration.” *Journal of Monetary Economics* 54 (Supplement 1):1–22.
- Hofbauer, Josef and William H. Sandholm. 2002. “On the Global Convergence of Stochastic Fictitious Play.” *Econometrica* 70 (6):2265–2294.
- Hosios, Arthur J. 1990. “On the Efficiency of Matching and Related Models of Search and Unemployment.” *The Review of Economic Studies* 57 (2):279–298.
- Khan, Aubhik and Julia K. Thomas. 2008. “Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Investment Dynamics.” *Econometrica* 76 (2):395–436.
- Klenow, Peter J. and Oleksiy Kryvtsov. 2008. “State-Dependent or Time-Dependent Pricing: Does It Matter for Recent U.S. Inflation?” *Quarterly Journal of Economics* 123 (3):863–904.
- Koop, Gary, M. Hashem Pesaran, and Simon M. Potter. 1996. “Impulse response analysis in nonlinear multivariate models.” *Journal of Econometrics* 74 (1):119–147.
- Lee, Hanbaek. 2026. “A Global Dynamic Nonlinear Solution Framework and the Repeated Transition Method.” Working paper, University of Cambridge.
- Maćkowiak, Bartosz, Filip Matějka, and Mirko Wiederholt. 2023. “Rational Inattention: A Review.” *Journal of Economic Literature* 61 (1):226–273.
- Maćkowiak, Bartosz and Mirko Wiederholt. 2009. “Optimal Sticky Prices under Rational Inattention.” *American Economic Review* 99 (3):769–803.
- Mangin, Sephorah and Benoît Julien. 2021. “Efficiency in Search and Matching Models: A Generalized Hosios Condition.” *Journal of Economic Theory* 193:105208.
- Mattsson, Lars-Göran and Jörgen W. Weibull. 2002. “Probabilistic Choice and Procedurally Bounded Rationality.” *Games and Economic Behavior* 41 (1):61–78.
- Matějka, Filip and Alisdair McKay. 2015. “Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model.” *American Economic Review* 105 (1):272–298.
- McFadden, Daniel. 1974. “Conditional Logit Analysis of Qualitative Choice Behavior.” In *Frontiers in Econometrics*, edited by Paul Zarembka. Academic Press, 105–142.
- . 1978. “Modelling the Choice of Residential Location.” In *Spatial Interaction Theory and Planning Models*, edited by A. Karlqvist et al. North-Holland, 75–96.

- . 1981. “Econometric Models of Probabilistic Choice.” In *Structural Analysis of Discrete Data*, edited by Charles F. Manski and Daniel McFadden. MIT Press, 198–272.
- Midrigan, Virgiliu. 2011. “Menu Costs, Multiproduct Firms, and Aggregate Fluctuations.” *Econometrica* 79 (4):1139–1180.
- Nakamura, Emi and Jón Steinsson. 2008. “Five Facts about Prices: A Reevaluation of Menu Cost Models.” *Quarterly Journal of Economics* 123 (4):1415–1464.
- . 2010. “Monetary Non-Neutrality in a Multisector Menu Cost Model.” *Quarterly Journal of Economics* 125 (3):961–1013.
- Oskolkov, Aleksei and Francesco Lippi. 2026. “Misallocation with Lumpy Investment.” Discussion Paper 21212, Centre for Economic Policy Research.
- Pissarides, Christopher A. 2000. *Equilibrium Unemployment Theory*. MIT Press, 2nd ed.
- Reis, Ricardo. 2006. “Inattentive Producers.” *Review of Economic Studies* 73 (3):793–821.
- Reiter, Michael. 2009. “Solving Heterogeneous-Agent Models by Projection and Perturbation.” *Journal of Economic Dynamics and Control* 53 (6):649–665.
- Rogerson, Richard. 1988. “Indivisible Labor, Lotteries and Equilibrium.” *Journal of Monetary Economics* 21 (1):3–16.
- Santos, Manuel S. 1991. “Smoothness of the Policy Function in Discrete Time Economic Models.” *Econometrica* 59 (5):1365–1382.
- Sims, Christopher A. 2003. “Implications of Rational Inattention.” *Journal of Monetary Economics* 50 (3):665–690.
- Stevens, Luminita. 2020. “Coarse Pricing Policies.” *The Review of Economic Studies* 87 (1):420–453.
- Thomas, Julia K. 2002. “Is Lumpy Investment Relevant for the Business Cycle?” *Journal of Political Economy* 110 (3):508–534.
- Tsallis, Constantino. 1988. “Possible Generalization of Boltzmann-Gibbs Statistics.” *Journal of Statistical Physics* 52 (1–2):479–487.
- Vavra, Joseph. 2014. “Inflation Dynamics and Time-Varying Volatility: New Evidence and an Ss Interpretation.” *Quarterly Journal of Economics* 129 (1):215–258.

Woodford, Michael. 2009. “Information-Constrained State-Dependent Pricing.” *Journal of Monetary Economics* 56:S100–S124.